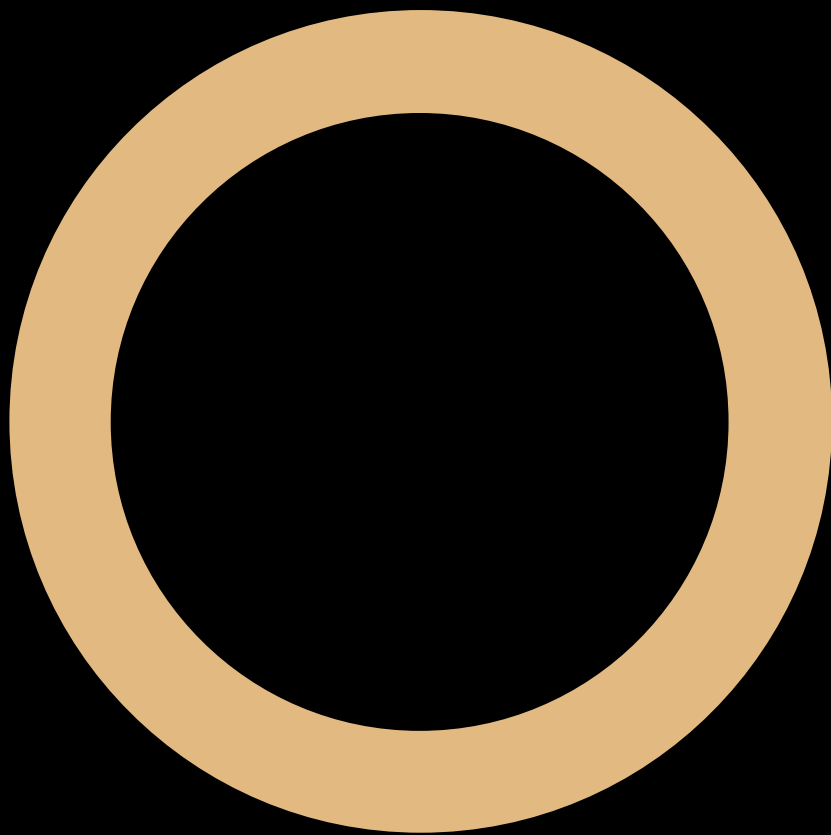




OSCAR
40

**OSG^{ar}
40**

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OSCar40

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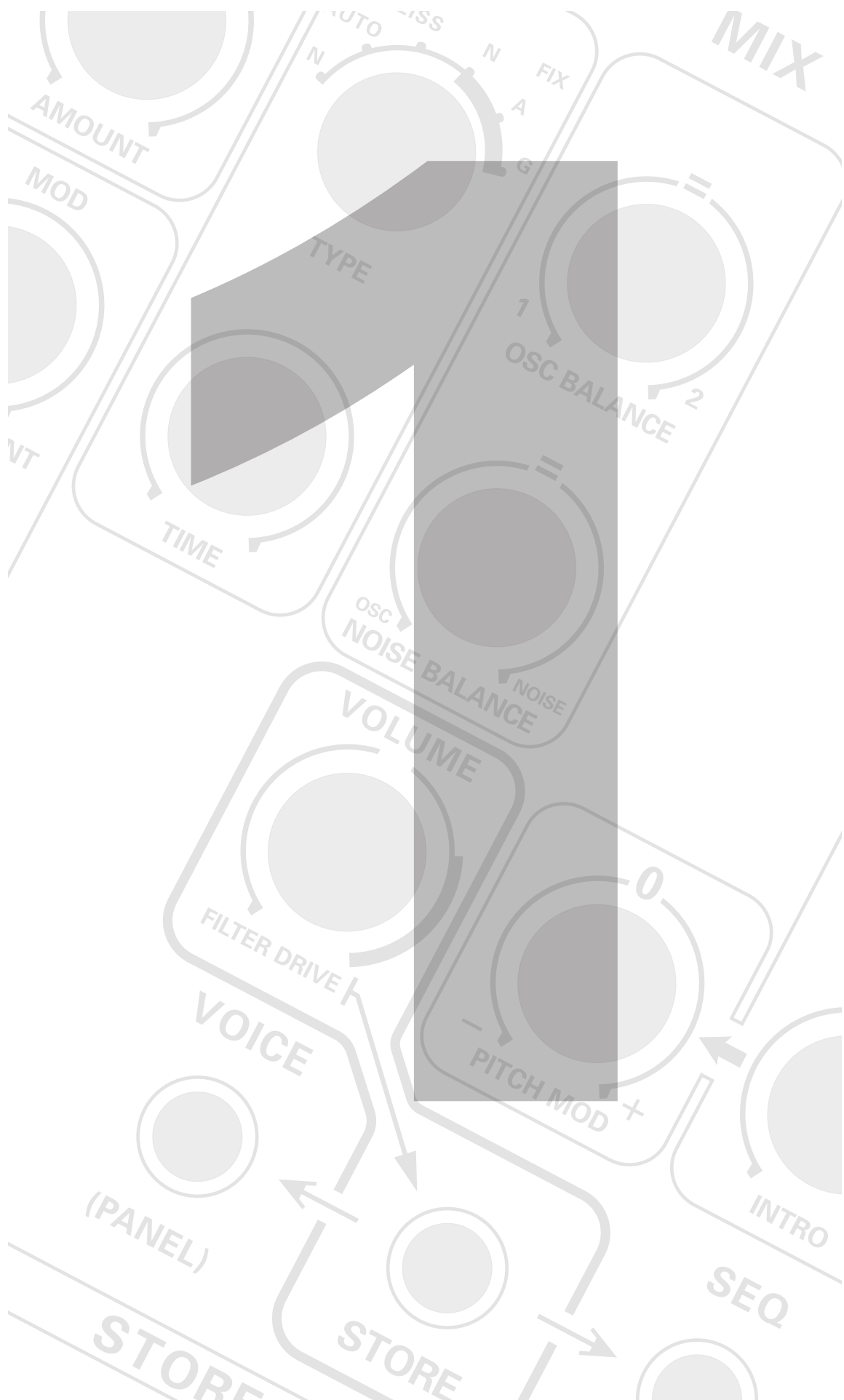
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1 INTRODUCTION

Meet the OSCar40

Thanks for your interest in the OSCar40. The OSCar40 is a 1:1 component-level recreation of the original OSCar mono synth designed by British synthsiser designer *Chris Huggett*, released in 1983 by the *Oxford Synthesiser Company*.

The OSCar40's design resolves many issues of the original while retaining all of its character and quirks.

The OSCar40 utilises a new operating system developed by Bob Grieb (Tauntek.com), with many additional features and a comprehensive MIDI specification including full MIDI CC control of all front panel controls.

This manual is similar in structure to the original OSCar manual, now updated with new feature descriptions, diagrams, and improved layout for clarity and readability.

I hope you enjoy playing your OSCar40 as much as I enjoyed building it.



The OSCar40 is a monophonic synthesizer and sequencer designed for both creative flexibility and reliable performance. By combining analogue components with digital control via a microprocessor, it provides powerful, high-resolution sound with modern convenience.

With quartz crystal-derived tuning, the OSCar40 maintains superb stability and accuracy, eliminating the need for auto-tuning and ensuring programs sound as originally set. Digital control removes parameter drift and reduces the need for constant manual adjustment.

A variety of advanced features—such as the arpeggiator, sequencer, and user waveform creation—are easily accessible with intuitive controls. All programs, sequences, and waveforms can be securely stored in memory, even when powered down, and can be exported to cassette tape for unlimited archiving and recall.

VOICES (patches) - 36 X 3 Banks of pre-programmed voices, 2 x user RAM banks of 36 voices each.

Parameter control position ***FIND EDIT*** display aids voice editing

DCOS - 2 with monophonic or duophonic assignment

WAVEFORMS - triangle, ramp, square, variable pulse

PWM - with 4 programmable rates

USER WAVEFORMS - by harmonic addition

PITCH CONTROL - 5 octave settings (pushbutton selection)

FINE TUNE - **OSC2** de-tune and **OSC2** octave shift

INTERVAL TRANSPOSE OSC2 only or both from -7 to +7 semi tones

GLIDE - normal, fixed time and automatic (portamento or glissando)

MIX - **OSC1/OSC2** balance, **NOISE/OSC** balance and filter drive

FILTERS - two filters can be configured as **24 DB/OCT** low pass or high pass or **12 DB/OCT** per side bandpass with variable bandwidth

FILTER RANGE - 16Hz to 16KHz with variable **Q** (two peaks)

LFO - for filter and pitch modulation

LFO WAVEFORMS - triangle, ramp, square and random plus env 2 or KYB

LFO RATE - 0.1 to 30 Hz

LFO MOD. DEPTH Pitch +/- 7 semitones, Filter +/- entire range

LFO INTRO DELAY - 0 to 8 sec after playing new key

JOYSTICK PITCH BEND - 1 octave up or down (max)

JOYSTICK MODULATION - filter/pitch modulation + or -
Separate filter and pitch amount controls.
Joystick modulation provides alternative LFO routing

ENV. GEN.1 - for VCA control - wide range ADSR with auto repeat

ENV. GEN.2 - for filter and pitch control - wide range ADSR with auto repeat
Alternative simplified envelope with trigger delay Modulation can be inverted

TEMPO - clock for envelope repeats, arpeggiator and sequencer playback

TEMPO RANGE - 6 to 1800 BPM

TRIGGER - single, multiple, env 1 and/or env 2 repeat
Automatic *gate time* facility *hold*

EXTERNAL TRIGGER OUT - with optional 1,2,3,4,6,8,12 or 16 divide

EXTERNAL TRIGGER IN - with optional 1,2,3,4,6,8,12 or 16 divide

ARPEGGIATOR - up, down, up/down or random at **TEMPO** rate
Choice of *HANDS ON* or *MEMORY* playing style

SEQUENCER - 22 independent sequences with up to 1500 events
Note tie and keyboard controlled legato
Repeat up to 24 times uses only 1 extra event space
Full edit functions (step up and down, insert, delete, replace)
10 sequences can contain chains of the other 12, with optional voice changes (as well as notes) Single or Loop playback

SYSEX DATA SAVE - gives unlimited permanent storage of voices, waveforms and sequences

MIDI CONTROL CHANGE support allows for automation from a DAW or MIDI controller.



What kind of synthesiser is OSCar40?

To answer this question it is best to think of it as three separate things in one enclosure:

1. A general purpose monophonic synthesiser with a wide range of controls for voice creation and playing.
2. A memory unit for saving 'sounds' which have been set up on the synthesiser so that they can be recalled instantly later on.
3. A sequencer with a comprehensive set of controls for creating and altering sequences of notes.

The Synthesiser

The synth section provides the expected control layout of a modern compact mono synth. OSCar40's sound palette stretches from classic analog to wild SFX, only limited by your experimentation. Performance features include precision transposing, stable tuning, modulation effects (auto or manual), octave selection, an arpeggiator, and **DUO** mode for limited polyphony.

The Memory Unit

The programmable memory lets you set up and save 36 user voices and 5 built waveforms. All are retained after power-off by battery backup. Voices are spread over 3 ROM banks with 36 each (total 108 presets), plus 2 RAM banks for your edits and creations. Preset or user voices can be freely modified and saved.

The Sequencer

The sequencer is built in—no external CV/trigger needed*. Store up to 22 sequences at once and freely edit, combine, or chain them (up to 10 can contain other sequences and voice changes). Repeats and chaining make creation of large structures efficient, and all sequence work is retained even after power-off.

*Sequencer steps *can* be triggered from a CV (+5V) trigger via the external trigger input. Arpeggiator steps can also be triggered in the same way.

Battery backed up memory

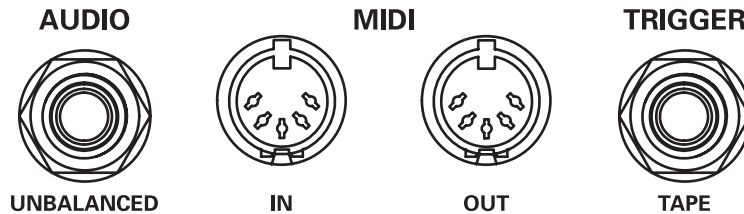
A CR2032 or BR2032 battery (accessed at the rear of the MIDI/Memory board) keeps memory alive between sessions. If you experience memory loss, check the battery. Always back up your data via regular MIDI SYSEX dumps to avoid loss.

The procedure to change the battery is described on the following pages.

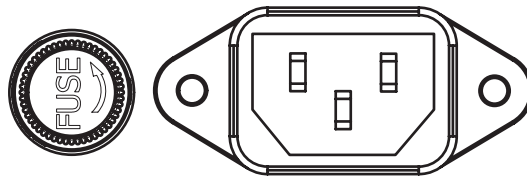


Rear Panel Connections

Left side upper (from rear) connectors



Right side lower (from rear) connectors



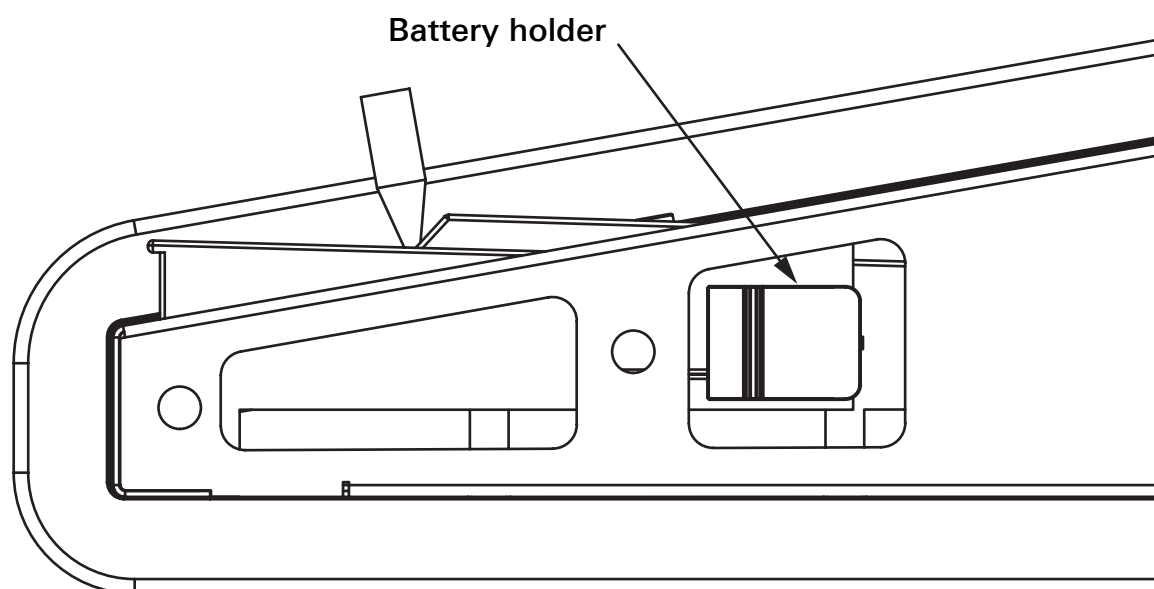
Replace the fuse with a **250V 2A 5x20mm Time Delay/Slow Blow** fuse. For OSCar40's with serial numbers ending in 25, locate the external fuse holder at the rear. For OSCar40's with no serial number, the fuse is located internally, located on the power supply circuit board. Follow the steps for disassembly described in battery replacement section.

Replacing the Memory Backup Battery

For OSCar40's with no serial numbers (manufactured in 2023/24), the battery is located on rear side of the MIDI circuit board located inside the enclosure. To access, the OSCar40 will need to be turned completely upside down, but be careful, you will need to place two long, solid objects, at least 5cm high, underneath both the left and right side cheeks to ensure that the joystick and knobs are protected. Once you have the OSCar40 upside down and secure, remove all the black M3 screws and place to one side.

***Do not remove the 6 M4 Torx screws, these hold the key-bed in place.
Do not try to remove the bottom panel immediately.***

With all the M3 screws removed, lift the front edge a few millimetres and carefully slide the bottom panel toward the rear by about 10 millimetres to clear the front edge of the keybed. You should then be able to lift the bottom panel higher, but not too far, you have to release the key-bed connector from the joystick circuit board first. You should be able to see and reach the grey ribbon cable, and follow it to the right where it connects to joystick board. Carefully loosen and unplug it. You can then lift the bottom panel and key-bed away from the main enclosure. The battery should be clearly visible on the left. Remove and replace the with a CR2032 or BR2032 battery. Reverse the steps above to reassemble, being careful to locate the key-bed and slide the bottom panel into place, keeping the front edge elevated to reconnect the key-bed connector to the joystick board.



For OSCar40's with serial numbers ending in 25 or later, the backup battery is located behind the right side end cheek. To access the battery holder, place the OSCar40 on a flat surface with easy access to the right side of the instrument. Don't turn it upside down. Place a folded towel or similar underneath the bottom panel of the OSCar40 sufficiently thick enough to hold the enclosure up when the side cheek is removed. Carefully remove the four M6 socket bolts and washers and place to one side, remove the side cheek. You will see the bare side of the enclosure, locate the black battery holder (maybe in a different orientation to diagram) and carefully lift the top flap. Remove the battery and replace with a new CR2032 or BR2032 battery. Reverse the steps above to replace the side cheek.





2 STARTING TO PLAY

If the OSCar40 initially seems complex, don't be discouraged. You don't need to master everything before starting. This section lets you start playing right away and introduces the instrument's main capabilities.

How the Controls Work

Most of the controls on the panel are used for forming different sounds and, to a large extent, the way they are used is fairly *normal*. However, the controls for selecting voices, transposes, sequences and waveforms require some explanation. The main thing to get used to is the idea of using the keyboard in conjunction with the push-buttons.

The simplest example of this scheme is the selection of a voice. Holding down the **VOICE** pushbutton tells the internal system that the next keyboard key that is pressed is to be interpreted as a voice setting *INSTEAD* of a note. The same idea applies to setting up an oscillator transpose, selecting a sequence, a special waveform or a harmonic.

Obviously, the keyboard is normally used for simply playing notes and none of these special functions need concern you if no push-buttons are held down. You can learn about each of them separately at the pace that suits you.



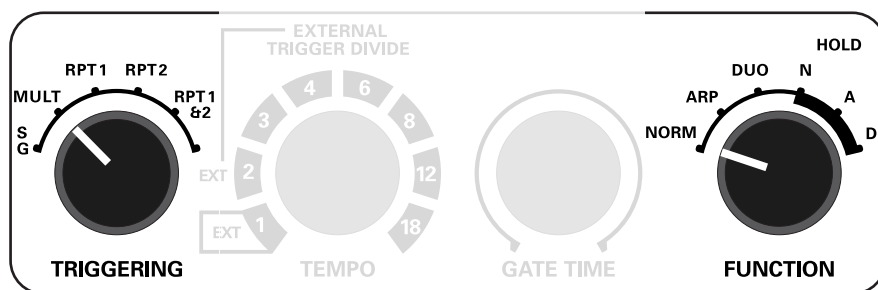
Switching On

When you switch on the OSCar40, you'll usually see the middle LED in the **OCTAVE** display light up—this is your indicator that the synth is powered and set to its central (default) octave.

The very last patch you used will automatically load if powering up for the first time (or after a battery change) **VOICE 1** from **ROM BANK 1** will be loaded.

Loading Voices

Before you select a **VOICE**, set these switches:



- **TRIGGERING** to **SG** or **MULT**
- **FUNCTION** to **NORM**



There are 36 programmable voices available from keys **-12** to **24**, excluding **0**, which is called the **PANEL VOICE**, more on this later*.

Loading a voice:

VOICE



1. Press and hold the **VOICE** button.
2. While holding, press any key between **-12** and **24** (key **0** is the **PANEL VOICE** – see below!)
3. Release both - your chosen voice is loaded.
4. Play notes to hear the sound

If you don't hear a sound check that the **TRIGGERING** and **FUNCTION** switches are set correctly.

What gets stored in a programmable voice?

The position of every control on the panel except **FINE TUNE**, **TEMPO**, the **VOLUME** control and the positions of the two wheels. Everything else is stored including the octave setting and both transpose settings.

***A note about VOICE 0 or the PANEL voice:**

Selecting **VOICE 0** uses all the current front panel positions, so what you see is what you get.

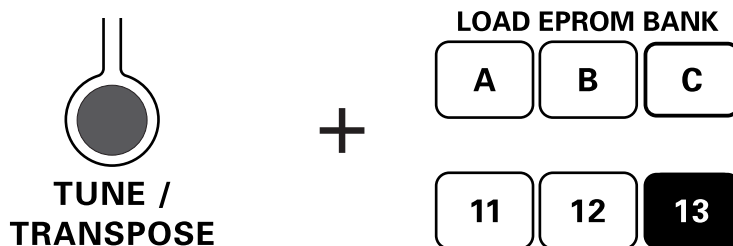
If you hear nothing or something unpleasant:

- Check your **VOLUME**
- Make sure **TRIGGERING** is set to **SG** or **MULT**
- Adjust your envelopes and waveforms as needed

PANEL VOICE can be a good starting point if you want to build a custom sound from scratch.

Loading the ROM Banks

There are three ROM banks available full of pre-programmed voices. Fresh from the factory ROM bank 1 is pre-loaded into working memory. The ROM banks are loaded by pressing and holding the **TUNE / TRANSPOSE** button



and then pressing either of the keys labelled:- **A (11)**, **B (12)** or **C (13)**.

Note that selecting a new bank does not immediately update the currently loaded voice. You have to load a voice from the new bank by pressing and holding the **VOICE** button and then pressing one of the keys.



Transposing the Oscillators

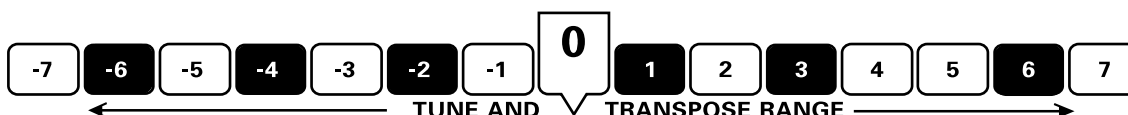
Interval transpose

INTERVAL offsets the pitch of oscillator 2 to oscillator 1's base frequency over a range of ± 7 semitones.



OSC 2
INTERVAL

Hold the **INTERVAL** button and press a key **-7** to **+7** to shift the pitch of **OSC2** relative to **OSC1** hold TUNE and use the same range to shift both oscillators.



Pressing the **INTERVAL** button and then releasing it *without pressing any keys* will reset the transposition.



TUNE /
TRANSPOSE

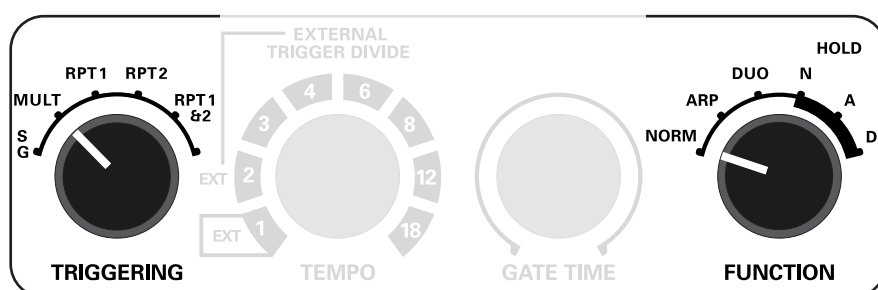
In a similar way, the **TUNE/TRANSPOSE** button can be used with keys in the transpose range to change the pitch of *both* oscillators. A transpose of both oscillators is called a *tuning transpose* since it effectively re-tunes the whole instrument.

Modifying (Editing) Selected Voices

When a pre-programmed voice is selected, you might notice the **OCTAVE** display blinking—this shows that the panel knobs are not set to match the loaded sound. This matters if you intend to tailor the voice further, as the actual sound settings often differ from the physical positions of the controls.

Holding **STORE** while turning any control and observing the **OCTAVE** display. LEDs will show which direction to turn until you find the stored position; once there, you have manual control. The **EDIT FIND** display appears only during this process, and practice will make it more intuitive.

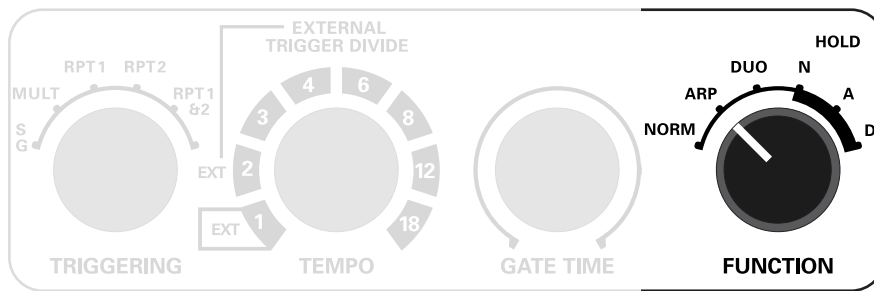
Start by using this feature to learn each knob's starting point, then move on to shaping the sound to your liking. For newcomers, this step-by-step workflow builds confidence and familiarity.



If you change knob or switch positions and the sound disappears, remember to return the **TRIGGERING** switch to **SG** or **MULT** and the **FUNCTION** switch to **NORM**.



ENVELOPES AND TRIGGERING



Using the Arpeggiator

To use the arpeggiator, select a voice and set the **FUNCTION** switch to **ARP**. Hold down several keys and use the **TEMPO** control to adjust the speed.

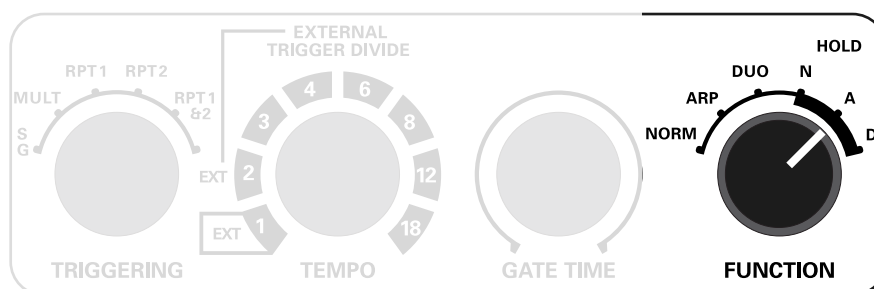
All held keys will play in sequence for as long as you hold them.

Try this with different preset voices. The arpeggiator works whenever the **FUNCTION** switch is set to **ARP**. Preset voices do not automatically engage **ARP**, but if the **FUNCTION** switch is set to ARP when selecting a voice, it will override the presets normal mode.

FUNCTION settings are stored with user voices so it's possible to load a voice and have the arpeggiator start immediately.

Arpeggiator Options

The way in which the arpeggiator works can be changed in several ways:



HANDS-ON mode

Only keys currently held down are played. To activate **HOLD** mode, turn the **FUNCTION** switch to **HOLD A**. This will keep the arpeggiation running even after you release the keys, though only one note will sustain when you let go.

INSERT

DELETE



MEMORY (INSERT) mode

Press and release the **INSERT** button in the **SEQUENCE EDITING** section. Now, as long as you hold any key, you can add more notes to the arpeggiation, and they will remain in memory until you clear it by pressing new keys. To return to *HANDS-ON* mode, press and release the **DELETE** button.

By default notes are stepped up and down the keyboard. This can be changed to *UP* only or *DOWN* only using the two **STEP** buttons in the **SEQUENCE EDITING** section or random by pressing and holding the **TUNE** button and key **8 / RND**.

The **STEP →** and **← STEP** buttons are used to change the direction modes:-

While holding a note on the keyboard, press and release the **← STEP** button and the arpeggiation will become **DOWN** only.

While holding a note on the keyboard, press and release the **STEP →** and the arpeggiation will become **UP** only.

To get back to the **UP/DOWN** mode, hold a key down (for the same reason as before) and press and release both **STEP** buttons together.

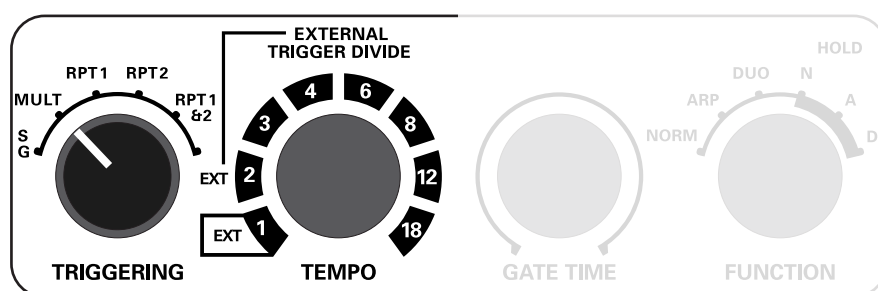
HANDS-ON or *MEMORY* and *UP*, *DOWN* or *UP/DOWN* or *RANDOM* settings are remembered when the voice is stored.

NOTE:

The **STEP →** button also starts sequence playback. To prevent this, always keep at least one key pressed when selecting arpeggiator *UP* mode. Although it may seem like a lot to remember, it's not an issue if a sequence accidentally starts—you can simply stop it using the **← STEP** button.



Different Types of Triggering



Most preset voices use the **MULT** (multiple triggering) position for the **TRIGGERING** switch. This means the envelope generators activate each time a new key is played, even if others are still held down—usually the most useful option.

You can select **SG** (single triggering) which only triggers the envelopes for the first key pressed.

As before, changing the type of triggering is just like any voice modification. Voice **6 (ROM BANK A)** is a good choice for demonstrating the effect of different types of triggering; it is a strong *plucked* sound which is highly dependent on the envelope generators for its character. Select voice **6** and play a few notes.

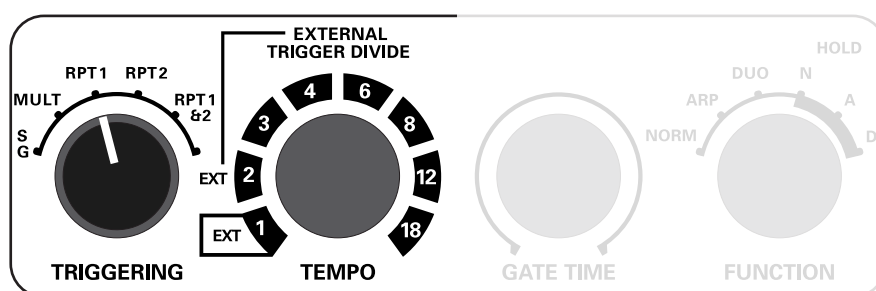
VOICE



(PANEL)

You will find that it always sounds the same how ever the keyboard is played. Now select **SG** on the triggering switch (in the usual way) and play again. Unless all the keys of the keyboard are released before playing a note, the strength of the sound is lost. By changing the way in which the keys are played, it is possible to control the attack quality of the note.

Envelope Repeats



You can set the envelope generators to repeat at the **TEMPO** clock rate.

With voice **6** selected, turn the **TRIGGERING** switch to **RPT 1** - this triggers the volume envelope (envelope 1) on every tempo beat. Hold a key and you'll hear the attack portion repeat steadily; adjust **TEMPO** to change the rate.

Switch to **RPT 2** for the filter envelope (envelope 2) to trigger on each beat, creating a different effect. Choose **RPT 1&2** to make both envelopes trigger with the tempo, simulating repeatedly pressing a key. This works because pressing a key always triggers both envelopes (except in single triggering if keys remain held).

Envelope repeats are great for creating regular rhythms that are hard to play manually, and can also be synced with other instruments like drum machines (see section 4 - *TRIGGER, GATE AND ENVELOPES*).

The **TRIGGERING** switch acts like the **FUNCTION** switch: it's programmable in **SG** or **MULT** mode only. In other positions, it overrides the selected voice's setting. To keep it programmable, return it to **SG** or **MULT** when selecting voices.

Envelope Repeats with the Arpeggiator

You may have noticed that the attack quality of many voices is lost when the arpeggiator is used to cycle through held notes. This happens because the arpeggiator itself doesn't trigger the envelope generators. However, you can restore the attack by triggering the envelopes automatically at the start of each arpeggiator note, using the envelope repeat function described earlier.

With voice **6** selected, activate the arpeggiator by turning the **FUNCTION** switch as before and test it out. While the arpeggiator is running, the sound's strength is only clear when you press new keys.

Turn the **TRIGGERING** to **RPT 1** to make the volume envelope generator trigger on each tempo beat.



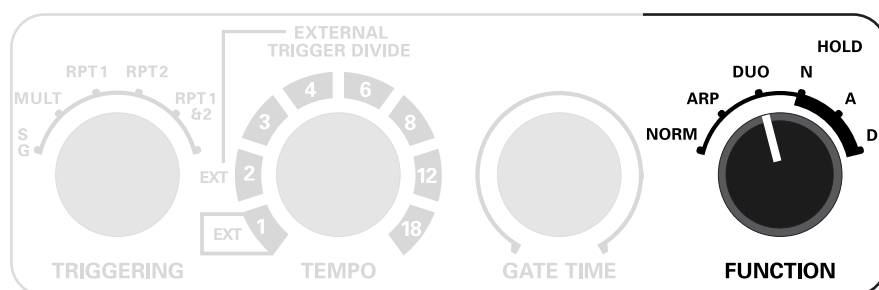
Now switch to **RPT 2** to make the filter envelope generator repeat and the effect is to restore the quality of the sound but in a different way.

Finally, switch to **RPT 1&2** to make both envelopes repeat.

Suppose that this set-up will be wanted during a live performance. All you need to do is store it in one of the **USER** voice locations while it is set up.

If this voice is selected later with the **TRIGGERING** switch at **SG** or **MULT** and the **FUNCTION** switch at **NORM**, the envelope repeat and arpeggiator functions will immediately come into operation.

Duophonic playing



Selecting duophonic mode is as simple as enabling the arpeggiator. With any voice selected, turn the **FUNCTION** switch to **DUO**.

For example, select voice **1**, switch to **DUO**, and play a few notes. With one key pressed, both oscillators play the same note; with two keys, each oscillator plays a different note. That's all there is to it.

The **DUO** setting can be stored, and will be active when that voice is recalled (with the **FUNCTION** switch in **NORM**).

Simple Sequences

The sequencer, despite its many controls, can be used simply:

1. Hold **SEQ** and **STORE** together, then press key **1**. This selects sequence **1** and enters editing mode. The **OCTAVE** display will reverse to show you're editing.
2. Press **INSERT** and **DELETE** together to clear sequence **1**.
3. Make sure that the **FUNCTION** switch is in the **NORM** position (or you will get confusing results). Play a few notes to make up a sequence.
4. To correct mistakes, clear the sequence and start again. Use the **SPACE** button for rests, which will be heard as *noise* during editing.

5. When finished, press both **STEP** buttons together to exit editing mode. The sequence is now ready to play.

Press the **STEP → / START/REPEAT** button to start the sequence playing. It will play through once only. Adjust the playing speed with the **TEMPO** control.

If you press the button again *WHILE THE SEQUENCE IS PLAYING*, it will make the sequence repeat in an endless loop.

Any notes which were played *legato* style (keys were played without releasing the previous key) will be played back in the same way.

Any voice can be selected in the usual way while a sequence is running. (This applies only to sequences **1** to **12** - see section **7 - The Sequencer** for more information.)

To stop the sequence playing immediately, press the **← STEP / SINGLE/ STOP** button twice. If you want to let the sequence run through to the end and stop, press the **← STEP / SINGLE/STOP** button once.

See section **7 - THE SEQUENCER** for more details on sequencer editing.

NOTE:

If the entire sequence space is full, you won't be able to create new sequences and while in *EDIT* mode playing the keyboard will give no response. To free up space, press **SEQ** and **STORE** together, then press key **2** to enter edit mode for sequence **2**. Clear it using **INSERT** and **DELETE**. Repeat with sequences **3**, **4**, **5**, and **6** if needed. Once you've cleared enough space, return to the previous steps.

Sequence Accompaniment

With a sequence running, you can accompany it on the keyboard by selecting **DUO** mode:

Start a sequence in an endless loop, select your voice, and set the **FUNCTION** switch to **DUO**. The sequence will play on oscillator 2, leaving oscillator 1 free for you to play manually. For more details on sequence accompaniment, see section **8 – THE DUO FUNCTION**.



Special Waveforms

The two **WAVEFORM** rotary switches in the **OSC1** and **OSC2** sections let you select the standard waveforms.

Additional waveforms can be chosen with the two **WAVEFORMS** push-buttons next to the transpose controls - just like selecting voices, but using the waveform buttons instead.

For example, select voice **21 (ROM BANK 1)** - a bright bass sound with sawtooth on oscillator **1** and pulse on oscillator **2**. To try different waveforms, hold the **OSC 1** button and press key **-3** to switch to an organ waveform. Repeat with the **OSC 2** button to change its waveform as well.

To change both oscillators at once, hold down both **OSC 1** and **OSC 2** buttons while pressing key **-3**.

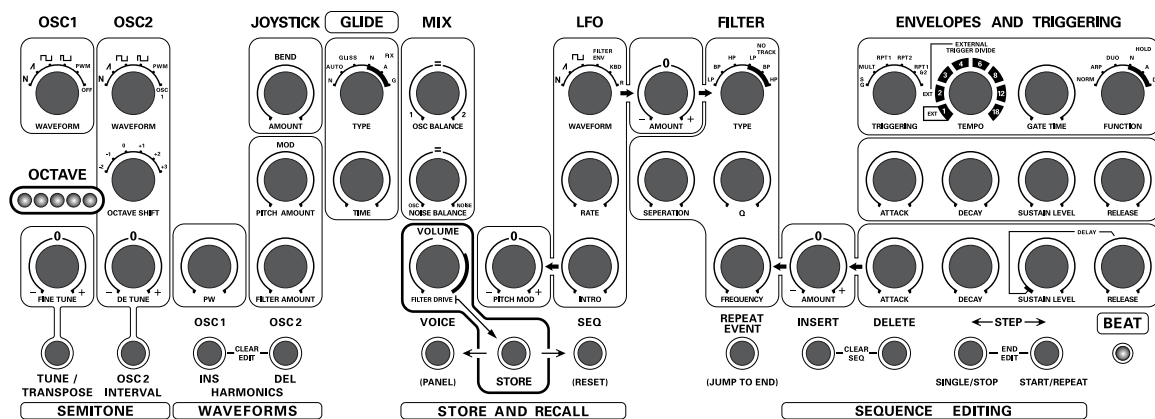
Getting the most out of the instrument

This section provides a solid introduction to the OSCar40, but doesn't cover everything. The rest of the manual explains all features in detail. If you're missing any information, check the relevant sections.

If you're unsure about anything, consider reading the manual fully before revisiting this section. It may clarify points you missed. Even if you feel confident, reviewing the manual could reveal useful tips you've overlooked.







3 THE CONTROL PANEL

This section describes all the controls on the panel, grouped by major parts of the system to help you understand how the instrument works. The control panel is divided into blocks, clearly labelled in large letters to help you find each control knob. References to sections or controls are written in **BOLD** for easy comparison with the panel.

Where no calibration marks are present, the minimum setting is fully anti-clockwise and the maximum is fully clockwise.

The following descriptions explain the function of each control, but not necessarily how to use them in every detail. Some controls are referenced in other manual sections for more information. Fully exploring their potential is part of the art of synthesizer programming and performance.

The Keyboard

The keyboard uses *new note priority*. This means that when any key is played, it will take play immediately even when other keys are already held down.

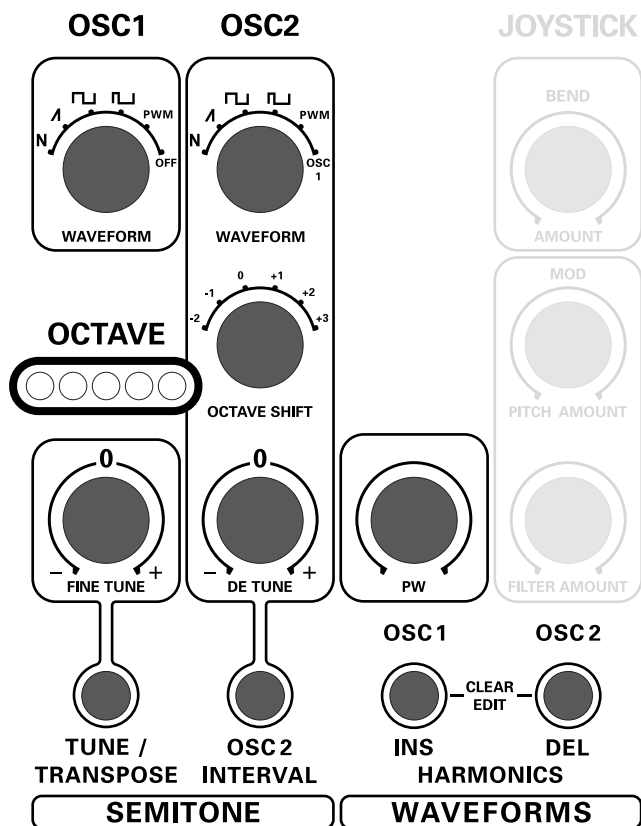
The keyboard is also used for many functions besides playing notes. Each key is numbered, from **-12** (bottom C) to **24** (top C), to help identify them when accessing other functions. Special keyboard functions are detailed throughout the manual.

The general process is: press and hold a control button (like **VOICE** or **SEQ**), press a key within the function range, then release both to activate the function.

When no buttons are held, the keyboard plays notes as normal—except during actions like cassette saving and loading or SYSEX dumps.



The Oscillators

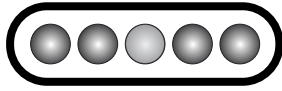


The two oscillators are nearly identical, and their pitches are usually controlled together - by the keyboard, sequencer, OCTAVE buttons, and other pitch controls.

The main exception is when using sequence accompaniment: oscillator 2 is then controlled only by the sequencer, disconnected from the keyboard, pitch bend, and modulation. Oscillator 1 remains fully responsive to the keyboard and pitch controls. For details, see the **DUO** section in section 6 – *ARPEGGIATOR, DUO & HOLD*.

Octave Display

OCTAVE



The five LEDs indicate the selected octave, as set by the OCTAVE buttons in the performance controls. The centre LED is the 8ft setting, with lower octaves to the left and higher octaves to the right. Standard footages are displayed accordingly:

LED	1	2	3	4	5
FOOT	32	16	8	4	2

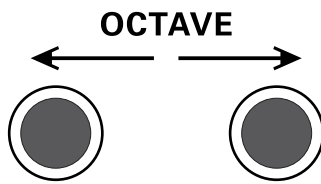
Usually, a single LED is lit to show the selected octave. However, in special cases, the LEDs display other information - typically with multiple LEDs lit, making it clear the display is not showing the usual octave setting.

The octave LEDs display other information in several situations:

- Sequence editing: The display reverses (four LEDs on, one off to show the selected octave).
- Voice selected: The display briefly blinks off about once per second to indicate a control is not set for the selected voice. Voice editing: Shows the effective position of a variable control (*EDIT FIND* display).
- SYSEX save/load: Indicates what's being transferred and any load errors.
- Other operations: Various blinks confirm actions have occurred. These are minor but provide positive feedback that a function has been executed.



Octave Controls



The **OCTAVE** push-buttons, used for performance octave control, are located above the joystick.

← OCTAVE and OCTAVE →

The octave *UP* and *DOWN* buttons shifts the octave setting of the oscillators each time they are pressed over a 5 octave range and is indicated on the **OCTAVE** display LEDs.

Fine Tune



FINE TUNE is the main tuning control, adjusting both oscillators up or down by up to 1 semitone. It is not affected by voice selection.

For larger pitch shifts, use the **TUNE/TRANPOSE** push-button.

Octave Shift



OCTAVE SHIFT affects only oscillator 2, setting its octave relative to oscillator 1.

In the **0** position, both are the same. Lower (**-1**, **-2**) and higher (**+1**, **+2**, **+3**) shifts are available. Once set, the octave difference is maintained even when using the **OCTAVE** buttons.

Oscillator Detune



DETUNE adjusts oscillator 2's pitch relative to oscillator 1 by up to 1 semitone either way. A small *dead* zone in the centre makes it easy to find the zero point. This control is ideal for achieving *phat* detune effects.

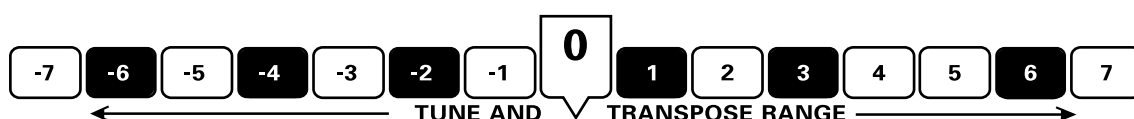
(For larger intervals, use the **INTERVAL** push-button.)

Oscillator Transpose and Interval



**TUNE /
TRANSPOSE**

The **TUNE/TRANSPOSE** push-button in the **SEMITONE** section shifts the tuning of both oscillators in exact semitones. To use, hold **TUNE/TRANSPOSE** and press any key between **-7** and **7** in the marked **TUNE AND TRANSPOSE RANGE**.



To reset tuning to key **0**, just press and release **TUNE/TRANSPOSE** without pressing any key or select Key **0** while holding **TUNE/TRANSPOSE**.

This gives standard tuning (A = 440Hz when **FINE TUNE** is in the centre). Using keys outside the marked range has no effect.



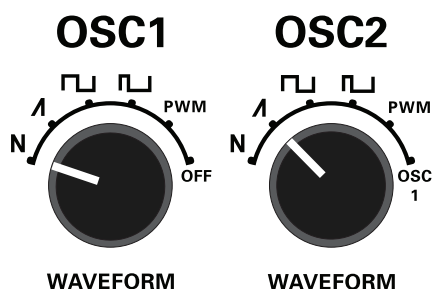
**OSC 2
INTERVAL**

The **INTERVAL** push-button works like the **TUNE/TRANSPOSE** button, but affects only oscillator 2.

Pressing and releasing **INTERVAL** without a key resets it to **0**.



Oscillator Waveforms



THE OSC1 and **OSC2 WAVEFORM** rotary switches allow independent selection of waveforms for each oscillator.

Position

- | | | |
|---|-------------|--|
| 1 | N | Triangular wave |
| 2 | A | Sawtooth wave |
| 3 | □ | Square wave |
| 4 | ▮ | Variable pulse width wave (width adjusted by PW) |
| 5 | PWM | Pulse Width Modulated wave (depth adjusted by PW) |
| 6 | OFF | OSC1 only - switches oscillator 1 off |
| | OSC1 | OSC2 only - causes oscillator 2 to adopt the same waveform as osc 1 |

About PULSE Waveforms

Variable pulse width and pulse width modulated (**PWM**) waves are similar to square waves, but the high and low portions of each cycle don't have to be equal. When equal (50% pulse width), the waveform is a standard square wave. With a narrow pulse width (e.g., 5%), the wave is high for only 5% of the cycle and low for 95%, creating a thin sound that matches the narrow shape.

Interesting sounds come from *changing* elements, and varying pulse width is no exception. With the variable pulse width wave, you can set the pulse width using the **PW** control; once set, it stays constant. However, as you adjust it, the sound becomes noticeably more vibrant. Using the **PWM** setting, you can continuously modulate the pulse width with an *LFO*, maintaining that lively, dynamic quality.

Pulse Width



The **PW** (pulse width) control is only active when either or both **WAVEFORM** select switches are set to position 4 or 5, and its function differs for each setting:

- **Position 4** (variable pulse width): **PW** adjusts the pulse width from 50% (square wave, fully anticlockwise) to a very narrow pulse (fully clockwise).
- **Position 5** (pulse width modulation): **PW** sets the amount of modulation above and below 50% pulse width. A dedicated LFO (one per oscillator) provides a triangular modulating signal; its rate changes according to the note played, with slightly different rates for each oscillator to optimise the effect. **LFO** and **TEMPO** are not involved. At the modulation peaks, the pulse width matches the variable pulse width setting.

Additional PWM rates



There are three additional **PWM** rates available, one faster than the control setting and two slower. Setting is stored in a voice.

They are accessed by:-

TUNE/TRANPOSE + key R1 (17)
sets the PWM rate back to its *slowest* setting

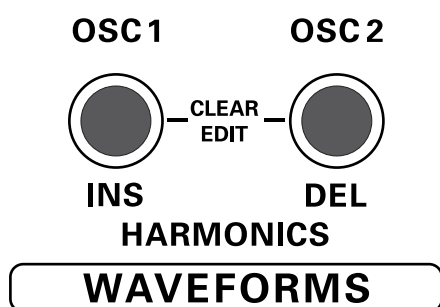
TUNE/TRANPOSE + key R2 (18)
sets the PWM rate to the *slower* setting

TUNE/TRANPOSE + key R1 (19)
sets the PWM rate to the *normal* setting

TUNE/TRANPOSE + key R1 (20)
sets the PWM rate to the *faster* setting



Special Oscillator Waveforms



The **OSC1** push-button in the **WAVEFORMS** section lets you change oscillator 1 to a special waveform using the keyboard.

Keys **-3** to **-7** are for preset special waveforms.

Keys **-8** to **-12** are for programmable waveforms which can be built up using the waveform *edit* facility.

Key **0** puts oscillator 1 into the waveform *edit* mode.

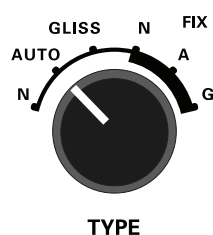
The **OSC2** push-button is functionally identical to that of the **OSC1** button except that oscillator 2 waveform is changed.

The **OSC1** and **OSC2** buttons can be used simultaneously to select the same waveform for both oscillators at the same time.

When one or both of the oscillators is in the waveform *edit* mode, the other markings around the buttons apply (**HARMONICS**, **INS**ert, **DE**lete and **CLEAR EDIT**).

See section **8 - SPECIAL WAVEFORMS** for a full description of special waveform selection, waveform building and storage.

Glide



The **TYPE** rotary switch in the **GLIDE** section lets you choose different glide types.

Position

- | | | |
|---|--------------|---|
| 1 | N | (normal) is standard portamento or smooth |
| 2 | AUTO | allows glide only if the last key played is still down |
| 3 | GLISS | (glissando) causes the glide to go in semi-tone jumps |
| 4 | FIX N | is normal portamento with fixed time (see note below about the fixed time option) |
| 5 | FIX A | is auto portamento with fixed time |
| 6 | FIX G | is glissando with fixed time |

Note: The **FIX** (fixed time) settings (positions 4–6) work like the first three settings, but the glide time across the keyboard remains the same for each **TIME** control setting.

Glide Time



The **TIME** control adjusts how quickly oscillator notes change in response to the keyboard or sequencer.

The range of adjustment depends on the type of glide selected:-

from *instant* to 6 seconds **PER OCTAVE** for glide types 1-3

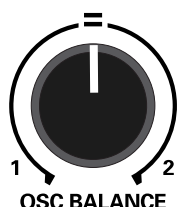
from *instant* to 1 second **OVER ANY NOTE SPAN** for glide types 4-6.



Sound Mix and Volume

The oscillator outputs are blended with the oscillator balance control, then mixed with the noise generator output using the noise balance control. The combined signal's level is set by the filter drive (volume) control before being sent to the filter.

Oscillator Balance



OSC BALANCE in the **MIX** section sets the level balance between both oscillators. In the centre position, both oscillators are at full level.

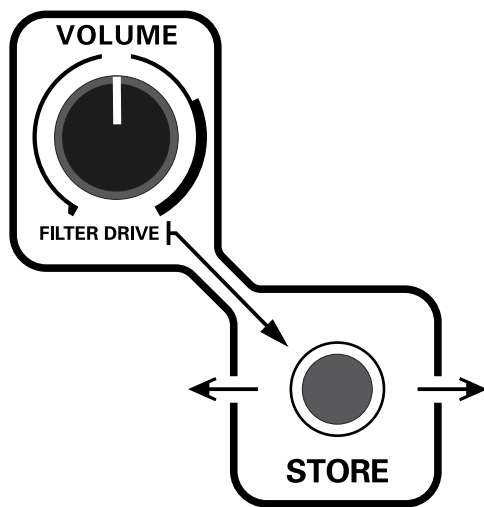
At **1** oscillator 2 is completely **OFF** and at **2** oscillator 1 is completely **OFF**.

Noise Balance



NOISE BALANCE adjusts the level between the oscillator mix (from **OSC BALANCE**) and the noise generator. In the centre, both are at full level. Turned to **OSC**, only the oscillators are heard; turned to **NOISE**, only noise is heard.

Volume



The **VOLUME** control has 2 functions. Normally it behaves simply as a volume control. But it also controls the *filter drive*.

While **VOLUME** is being turned it will show the *EDIT FIND* display for its second function:- the *filter drive* and is showing you where the filter drive is set. (See *STARTING TO PLAY* or *VOICES* for an explanation of the *EDIT FIND* display.)

To adjust filter drive: Hold the **STORE** button (as shown on the panel arrow), which keeps the volume steady while you use the **VOLUME** knob to change filter drive. The *EDIT FIND* display helps locate the setting. When you release **STORE**, the filter drive is set, and the knob returns to normal volume control. The output will immediately jump to the current knob position.



Pitch Bend and Modulation

The main modulation source is the LFO (low frequency oscillator), which can modulate oscillator pitch and filter cutoff frequency. The intensity of the LFO's effects can be controlled in two ways:

1. ***Automatic intro delay:*** Modulation fades in after an adjustable delay when a key is pressed. The amount of delay is set using the **DELAY** control. It is independent of the joystick.
2. ***Using the joystick:*** Moving the joystick up/down on the **MOD** axis applies modulation to pitch and filter, with programmable amounts for each. This is independent of the **DELAY** setting.

Modulation amounts for pitch and filter from each source are fully independent and can combine either positively or negatively.

Pitch bend is separate - it shifts oscillator pitch directly, using the **BEND** axis of the joystick, which also has a programmable amount control.

The following section explains the **LFO** controls, then the intro delay and joystick routing controls.

LFO



The **WAVEFORM** rotary switch in the **LFO** section selects the type of signal the **LFO** produces. Positions labelled **KBD**, **ENV**, and **R** aren't true **LFO** waveforms but offer convenient routing for additional effects.

Position

- | | | |
|---|------------|---|
| 1 | N | triangular wave |
| 2 | A | sawtooth wave |
| 3 | □ | Square wave modulation only moves in one direction (set by the amount controls). This is ideal for trills, as it keeps the basic pitch unchanged.. |
| 4 | ENV | (envelope) selects the filter envelope instead of an LFO waveform |
| 5 | KBD | (keyboard) selects the equivalent of the keyboard control voltage This lets you adjust keyboard pitch tracking or filter tracking from their normal settings. |
| 6 | R | (random) uses a sample-and-hold of the noise generator, updated at the start of each LFO cycle and held until the next. Adjust the RATE control to set how often new random levels occur. |

LFO Rate



The **RATE** control sets the **LFO** frequency from 1 cycle every 30 seconds up to 30 cycles per second.



LFO Intro



INTRO sets a delay before modulation begins when a new keyboard note is played; modulation (on **PITCH MOD** or **AMOUNT** to filter) starts at zero and fades in. Delay ranges from zero to about 8 seconds. The triggering mode **SG** (single) or **MULT** (multiple) also affects how **INTRO** resets.

LFO Pitch Modulation

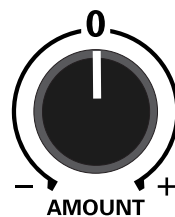


This control does not have any effect on the joystick **MOD**ulation settings.

PITCH MOD sets how much the **LFO** modulates oscillator pitch. In the centre (0), modulation is off, with a small dead zone for easy zeroing.

Turn clockwise for positive modulation (e.g., rising sawtooth or upward square wave), and anticlockwise for a negative effect. Maximum range is a 7 semitone shift.

LFO Filter Modulation



This control is independent joystick settings, but is affected by the **INTRO** delay.

The **AMOUNT** control sets how much the **LFO** modulates filter frequency, much like **PITCH MOD** affects pitch. Maximum modulation covers the filter's full range.

OSC1 LFO Modulation

LFO - OSC1



LFO modulation to **OSC1** can be disabled/enabled by:-

TUNE/TRANSPOSE + key **23 (ON)** then release
enables LFO modulation of **OSC1**

TUNE/TRANSPOSE + key **24 (OFF)** then release
disables LFO modulation of **OSC1**

Setting saved in a voice.

LFO Sync Reset

LFO SYNC



LFO waveform can be reset when the Arpeggiator or Sequencer clocks to the next note. This setting is saved in patch. It is normally **OFF**.

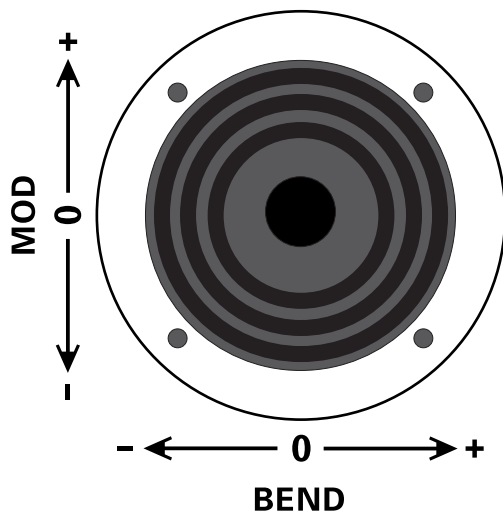
Having **LFO SYNC** enabled without using the Arpeggiator or Sequencer, i.e. normal play, can yield some undesirable results, as the LFO will reset every **BEAT** at the rate set by the **TEMPO** control, potentially disrupting smooth LFO modulation effects.

TUNE/TRANSPOSE + key **21 (OFF)** then release
enables LFO sync

TUNE/TRANSPOSE + key **22 (ON)** then release
enables LFO sync



Joystick



The joystick (to the left of the keyboard) is the *performance* control for altering pitch and modulation effects while playing. The joystick is self-centering and in the centre position it has no effect.

Joystick **BEND** axis (*left and right*)

This raises or lowers the oscillator pitches to an extent programmed by **BEND AMOUNT** control.

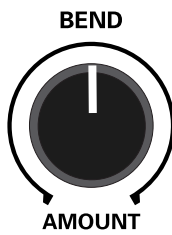
Joystick **MOD** axis (*up and down*)

This simultaneously controls the amount of **LFO** modulation of:-

1. **PITCH** programmed by the **JOYSTICK / PITCH AMOUNT**
2. **FILTER FREQUENCY** programmed by the **JOYSTICK / FILTER AMOUNT**

It is not affected by the **INTRO** delay or the alternative **PITCH MOD** and **FILTER AMOUNT** controls. Joystick modulation can be mixed in with **INTRO** modulation if required.

Bend Amount



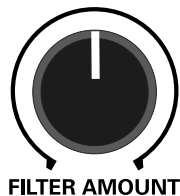
The **BEND / AMOUNT** control sets how much the joystick bends pitch along the pitch axis (left or right) anywhere from zero up to about 14 semitones in either direction at the joysticks extremes.

Pitch Mod Amount



The **MOD / PITCH AMOUNT** control adjusts the extent of the *LFO pitch modulation* obtained by moving the joystick along the **MOD** axis (*up or down*) from zero to a 7 semi-tone swing when the joystick is pushed to its extreme (positive or negative).

Filter Mod Amount



The **MOD / FILTER AMOUNT** adjusts the extent of the **LFO** filter modulation obtained from pushing the joystick along the **MOD** axis from zero to a swing over the full range of the filter when the joystick is pushed to its extreme along the **MOD** axis (positive or negative).



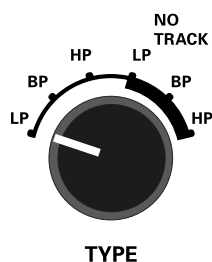
The Filter

The filter section uses two 12 dB/octave filters with *linked* cutoff frequencies. You can change the relationship of the cutoff frequencies with the **SEPARATION** control, letting you combine the punch of a 24 dB/octave filter with the smoother tones of a 12 dB/octave filter. **Q** control works as expected with no separation, but combining high **Q** with separation creates new sounds—especially in *bandpass* mode, where vocal-like effects are possible.

About Filter Tracking

Filter cut-off frequency usually tracks the oscillator's pitch, keeping filtering consistent as you play different notes - generally the most useful approach. However, you can set filter tracking to anything from over twice normal to negative values for special effects. To adjust this, select **KBD** on the **LFO** waveform switch and set a desired **LFO** filter amount. Direct filter **AMOUNT** routing is recommended (with **INTRO** delay at zero), as **MOD / FILTER AMOUNT** via joystick would require holding the joystick off-centre constantly.

Filter Type



FILTER / TYPE (of filtering) rotary switch allows the selection of the filter types and variations.

LP - low pass

Frequencies below the filter's cut-off pass through to the output, while higher frequencies are increasingly reduced - the higher the frequency, the less gets through. A steep 24 dB/octave slope (*with both filters in low pass and no separation*) greatly reduces high frequencies compared to a gentler 12 dB/octave slope.

HP - high pass

In high pass mode, frequencies above the cutoff pass through while those below are reduced. With both filters set to high pass and no separation, you get a steep 24 dB/octave slope.

BP - band pass

In this mode, one filter is low pass and the other is high pass, so only frequencies between their cutoffs are allowed through. Frequencies outside this band are reduced at 12 dB/octave. The **SEPARATION** control adjusts the bandwidth. When separation and **Q** are increased, the filters' resonant peaks move apart, creating distinct effects.

Positions 4–6 (**NO TRACK**) function like the first three filter settings, but the filter does not track oscillator **1**. If needed, you can still use the **KBD** setting on the **LFO** waveform switch for adjustable filter tracking.

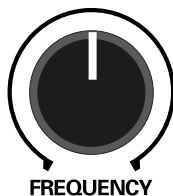
Filter Separation



The **SEPARATION** control sets the difference between the two filter cut-off frequencies, from zero (fully anticlockwise) up to about 4 octaves (fully clockwise). In **LP** and **HP** modes, zero separation combines both filters as a single 24 dB/octave filter. Increasing separation decreases the slope toward 12 dB/octave.

In **BP** mode, increasing separation increases the bandwidth of the filter.

Filter Frequency

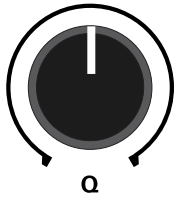


The **FREQUENCY** control manually sets the filters' cut-off frequency, covering the full audio range (under 16 Hz to over 16 kHz).

Both filters move together, maintaining any set separation. **LFO**, envelope, and tracking effects are applied on top of this setting.



Q - Resonance



The **Q** control sets resonance for both filters near their cut-off frequencies. With **Q** fully anticlockwise, there's no resonance. As **Q** increases, frequencies near the cut-off are boosted, creating a resonant effect. At maximum **Q**, the filters can even oscillate, making a tone at their cut-off frequencies even if the oscillators are off.

High **Q** combined with modulation or envelope control of the cutoff frequency creates strong, dramatic frequency sweeps (e.g., classic **WAH-WAH** effects). With two filters, setting separation gives two resonant peaks—especially noticeable in band pass mode.

Filter Envelope Amount



The **AMOUNT** control sets how much the filter envelope (envelope generator 2) modulates filter frequency—ranging from zero (with a small centre dead zone) to the filter's full range in either direction.

Positive (clockwise) settings make the envelope sweep *above* the manual **FREQUENCY** setting; **negative** (anticlockwise) settings sweep *below*. Set **FREQUENCY** to an appropriate starting point for best effect.

The Envelope Generators and Triggering

There are two envelope generators, both nearly identical. Each has four controls: attack, decay, sustain level, and release. Envelope generator 1 (the upper panel controls) is connected to the **VCA** (voltage controlled amplifier) and controls overall volume.

Envelope generator 2 mainly controls filter cut-off frequency, as indicated by the panel arrow. It can also be routed (via the **LFO** waveform switch) to create transient pitch effects for the oscillators if needed.

The two envelope generator sets aren't labelled *volume* and *filter*, but arrows on the panel indicate the lower set is for the filter envelope.

Envelope generators are usually triggered by playing the keyboard, but can also be triggered in other ways - including with the **TEMPO** clock for repetitive effects. For more on their operation, see section 4 – **TRIGGER, GATE AND ENVELOPES**. The next sections focus on the basic control functions.

Envelopes and Triggering

Volume Envelope Generator



ATTACK

The **ATTACK** control adjusts the time taken for volume to rise from zero to its full level (for any given volume control setting), after a trigger has occurred and while the *gate* signal is present. The range of adjustment is from less than 1 millisecond to over 15 seconds. The volume rises linearly (unlike the decay and release which are exponential).



DECAY

The **DECAY** control adjusts the time taken for volume to decay from its full level (the level reached in the attack phase) down to zero, directly after the attack has finished as long as the *gate* signal is present. The range of adjustment is from less than 1 millisecond to over 60 seconds to complete the full change in volume. If sustain level is set above zero, the decay will stop when the sustain level is reached, thus shortening the decay period.





SUSTAIN LEVEL

The **SUSTAIN LEVEL** control sets the volume level at which decay stops. This level is maintained while a key is held down *or* while automatic *gate time* is operating (ie while the *gate* signal is present).



RELEASE

The **RELEASE** control adjusts the time taken for volume to fall from the sustain level back down to zero. The range of adjustment is from less than 1 millisecond to over 60 seconds to cover the full volume change (from maximum to zero). If the release starts at less than the maximum level, the release time is correspondingly shortened.

The release starts after (all) the keys have been released *or* after automatic *gate time* has finished - if it is operating (i.e. when the *gate* signal disappears).

Filter envelope generator

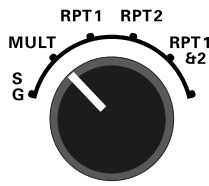
The four controls have the same effect on filter frequency as those of the volume envelope generator have on volume except for the following differences:-

ATTACK is exponential like the *decay* and *release* and the range of adjustment is from less than 1 millisecond to over 60 seconds.

If **SUSTAIN LEVEL** is set to zero (fully anticlockwise), there is an adjustable **DELAY** period between triggering and the start of the attack. In these circumstances, the **RELEASE** control is used to set the **DELAY** time instead of having its normal function. The range of adjustment of the delay is from zero to 1 second.

Note that if all keys of the keyboard are released *or* if automatic gate time finishes (*i.e. gate disappears*) before the **DELAY** is over, the attack will not start.

Triggering, Tempo, Gate Time and Function



TRIGGERING

The positions of the **TRIGGERING** rotary switch that would normally be used are **SG** for single triggering or **MULT** for multiple triggering of the envelope generators. In these two positions, both envelope generators are triggered together.

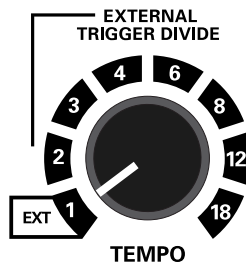
Also, in these two positions, the **TRIGGERING** function is programmable. In other words when a voice is selected, the effective position of the switch is set to whatever was programmed in the voice memory. In the remaining four positions it is **NOT** programmable; the switch position will over-ride the effective position that the selected voice contains. However, if a voice is stored, the position of the switch will always be memorised and will take effect if the voice is selected with the switch in either the **SG** or the **MULT** position.

In the three **RPT** (repeat) positions, either or both of the envelope generators is triggered by the **TEMPO** clock:-

- RPT1** the volume envelope is triggered by the **TEMPO** clock
- RPT2** the filter envelope is triggered by the **TEMPO** clock
- RPT1&2** both envelopes are triggered by the **TEMPO** clock

In the **EXT** position, both envelopes are triggered by external trigger pulses.

Also, the **TEMPO** knob acts as an **EXTERNAL TRIGGER DIVIDE** control as indicated above the knob. The choice of dividing factors is marked:- **1,2,3,4,6,8,12** or **16**.



The **TEMPO** controls the speed of repeated envelope triggering, arpeggiation and sequence playback. The range of adjustment is from about 1 beat every 4 seconds to 32 beats per second.

In the fully anticlockwise position marked **EXT**, the tempo clock stops operating and the **EXTERNAL TRIGGER IN** jack socket connection becomes active. All the functions which would normally be governed by the tempo clock are now operated from external trigger pulses.



This is simply an alternative way of obtaining external triggering but without restricting the **TRIGGERING** switch to the **EXT** position. Obviously it prevents the use of external trigger dividing.

If the **TRIGGERING** switch is set to **EXT**, the **TEMPO** control becomes an **EXTERNAL TRIGGER DIVIDE** control as explained above.

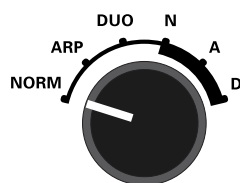
The **GATE TIME** control is active during envelope repeats and sequence playback. Its main purpose is to simulate the *key held down* time of normal playing during sequence playback. The range of adjustment is from 30 milliseconds to 8 seconds.

BEAT



The **BEAT** LED comes on at each **TEMPO** clock pulse or, if external triggering is being used, at each trigger pulse. It and stays on during automatic **GATE TIME**.

Function - Play Modes



FUNCTION

The **FUNCTION** rotary switch is used to select the play mode of the synthesiser.

NORM (normal) is used for normal monophonic playing. With the switch in this position, **FUNCTION** is programmable (when a voice is selected).

ARP (arpeggiator) brings the arpeggiator into operation at any time except when a sequence is running.

DUO (duophonic) changes the oscillator keyboard assignment so that two notes can be played simultaneously.

Positions **4** to **6** are the same as the first three positions except that an indefinite **HOLD** is provided *without affecting triggering*. The effect of **HOLD** is to simulate the effect of holding down a key of the keyboard.

It would be used for example to leave the arpeggiator playing in the **MEMORISE** mode while another instrument is being played.

In the **NORM** position, the effective position of this switch is programmable.

In other words the **FUNCTION** can be altered by selecting a voice. In any of the other positions, **FUNCTION** is *NOT* programmable.

The actual position of the switch over-rides voice memory settings, like the **TRIGGERING** switch in any positions other than **SG** or **MULT**. If a voice is stored, the position of the switch will always be included in the stored voice but will only take effect if the switch is in the **NORM** position when the voice is selected.

See section *6 - ARPEGGIATOR, DUO & HOLD* for more information about this switch.

Store and Recall and Sequence Editing

All the functions of these remaining controls are described in detail in other sections. Most of the controls have several different functions and they are all briefly listed here.

STORE AND RECALL

The **STORE AND RECALL** section is used for saving waveforms and voices and loading voices and sequences. *Note that loading ROM banks and USER banks is managed elsewhere.*

The **VOICE** push-button is used in conjunction with the keyboard to select and store voices. Its use is fully described in section *5 - VOICES*.

The **STORE** push-button is used in conjunction with other buttons for saving voices and waveforms and for starting a sequence edit. Its use is described in the appropriate sections.

It is also used while **FILTER DRIVE** is being adjusted (see *ADJUSTING FILTER DRIVE* in the **MIX** section).

STORE + SEQ (sequence) push-button
is used in conjunction with the keyboard to select or edit sequences.
See section *7 - THE SEQUENCER*.



SEQUENCE EDITING

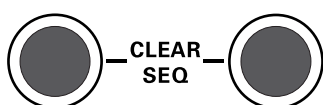
REPEAT EVENT



(JUMP TO END)

REPEAT EVENT push-button is used in sequence edit with the keyboard to enter repeats into sequences. It can be used on its own to jump to the end of the sequence being edited.

INSERT DELETE



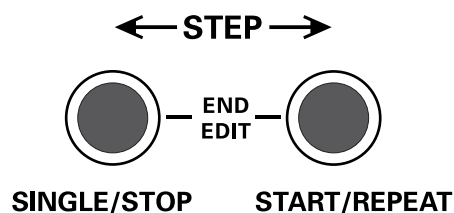
The **INSERT** push-button is used during sequence editing in combination with sequence entry commands to **ADD** events to the existing sequence. It is also used with the **DELETE** pushbutton to clear the sequence being edited.

The **INSERT** push-button in combination with other buttons is also used to initiate saving SYSEX data. For a complete explanation of saving and loading SYSEX data see section **9 - MIDI FUNCTIONS - Saving and loading SYSEX data**.

The **INSERT** button is also an **ARPEGGIATOR** mode control, used to select the **MEMORY** mode.

The **DELETE** push-button is used during sequence editing to delete one event from the sequence **OR** with the **INSERT** button as mentioned above.

It is also an **ARPEGGIATOR** mode control, used to set the **HANDS-ON** mode.



The **← STEP** (step down) push-button is used during sequence editing to step back through the sequence *OR* with **STEP →** to end the edit.

In sequence repeat, it sets playback to **SINGLE**.

In sequence single playback, it **STOPs** the sequence.

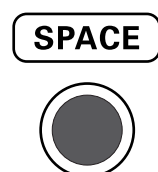
It is also an **ARPEGGIATOR** mode control, used to select direction of play.

STEP → (step up) push-button is used during sequence editing to step forward through the sequence *OR* with **← STEP** to end the edit.

In sequence standby, it **STARTs** a single play of the sequence from the beginning.

In sequence single playback, pressing it again while the sequence is playing sets indefinite **REPEAT** playback. A quick double press will start play and **REPEAT**.

It is also an **ARPEGGIATOR** mode control, used to select direction of play.



The **SPACE** pushbutton (to the left of the keyboard) is used during sequence editing to enter rest periods into the sequence.

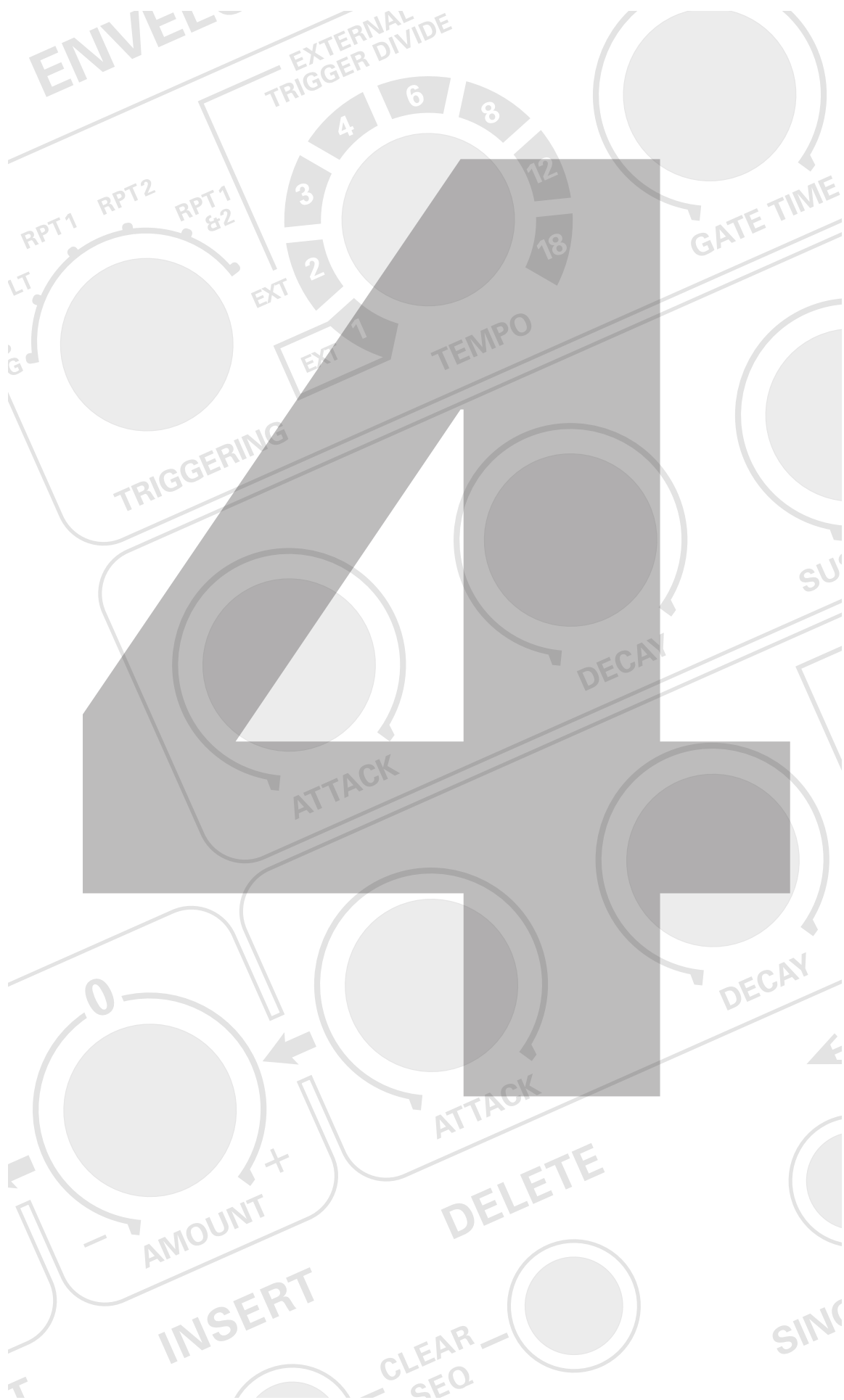
It can also be used when the **ARPEGGIATOR** is running in the **MEMORY** mode while inserting notes into the memory.

It is also used to initiate SYSEX dumps after the data type to be saved is selected.

For a complete explanation of sequencer functions, see section **7 - THE SEQUENCER**.

For a complete explanation of arpeggiator functions, see: section **6 - THE ARPEGGIATOR, DUO & HOLD**





4 TRIGGER, GATE AND ENVELOPES

General Explanation

For many synthesizer users, triggering and gating are straightforward concepts. However, since the *OSCar40* offers several, including some unique options, this section begins by explaining their sources and functions.

A **TRIGGER** is a *brief* event that starts the **ATTACK** phase of the envelope generators and usually causes the oscillators to switch to a new note, whether from the keyboard, arpeggiator, or sequencer.

The trigger signal can come from the *keyboard*, the *internal tempo clock*, or an *external* source like a drum machine or another synthesizer (via the external trigger socket—see *EXTERNAL TRIGGERING*).

The **GATE** is an *ON/OFF* signal that controls envelope generator activity after the trigger starts the attack phase. *The gate operates internally and is worth noting mainly to understand how the system works.*

When the trigger comes from the *keyboard*, the gate stays *on* as long as any key is held. If triggered by the *tempo clock* (during sequence playback or envelope repeats), the gate stays *on* for a duration set by the **GATE TIME** control - simulating key hold time. The **BEAT** lamp lights during this automatic gate period.



Envelope Generators

There are two envelope generators: one (upper group) controls the volume, and the other (lower group) controls filter cut-off frequency.

Both envelope generators operate similarly. After receiving a trigger, the sequence of events follows the control panel layout from left to right:

ATTACK is the rate at which full volume is reached (or the filter frequency fully sweeps).

DECAY is the rate at which the level drops back to the sustain point.

SUSTAIN LEVEL is the level that persists as long as the gate is *on*.

RELEASE is the rate at which it falls back to zero after the gate goes *off*.

If the gate turns *off* during the attack or decay phase, the release phase starts right away if the envelope level is below sustain. If it's above sustain, it depends on the decay and release rates: the faster rate takes over. If release is faster, release begins immediately. If decay is faster, the envelope drops to sustain at the decay rate, then falls to zero at the release rate. This setup works best in most situations.

The filter envelope has an extra feature: if the sustain level is set to zero, the release control instead provides a pre-attack **DELAY** (the time between the trigger and the attack phase). This allows for effects like a delayed filter sweep (echo-like effects). The release knob sets the length of this delay.

With delay active, the envelope output always drops at the **DECAY** rate regardless of the gate. If the gate turns *off* before the delay ends, the attack phase *won't* start.

The Tempo Clock

This is an independent oscillator that generates trigger pulses for timing the arpeggiator and sequence playback.

TEMPO sets the clock speed (unless fully anticlockwise - see **EXTERNAL TRIGGERING**). The **BEAT** LED flashes with every pulse, even when the tempo clock isn't in use.

NOTE: If *gate time* exceeds the tempo clock period, the **BEAT** lamp will stay lit continuously.

Trigger Selection

The trigger and gate signals for the volume envelope, filter envelope, and note changes can come from *different* sources. For example, the keyboard might control notes and volume, while the filter envelope is triggered by the tempo clock or an external trigger.

The **TRIGGERING** select switch lets you choose between the most practical and useful combinations of trigger sources, even though many different combinations are possible.

The typical **TRIGGERING** switch settings are **SG** (single) or **MULT** (multiple) triggering.

In **SG** mode, both envelope generators trigger from the keyboard *only* when a key is pressed and no other key is already held down.

MULT is similar to **SG**, except that a trigger occurs every time any key is played, regardless of whether other keys are held down.

In the other four positions, one or both envelope generators are triggered by the tempo clock or an external trigger, typically for special effects.

RPT1 triggers the volume envelope generator (envelope 1) on every **TEMPO** clock pulse while a key is held down. The filter envelope continues to trigger from the keyboard, using multiple triggering as described above.

RPT2 triggers the filter envelope generator (envelope 2) on every **TEMPO** clock pulse, regardless of other conditions. The volume envelope continues to trigger from the keyboard, with multiple triggering as above.

RPT1&2 triggers *both* envelope generators with the **TEMPO** clock, combining the effects of **RPT1** and **RPT2**.

In **EXT** mode, both envelope generators are triggered by external trigger pulses. This works like **RPT1&2**, but uses external triggers instead of the **TEMPO** clock.

When using repeat modes (**RPT1**, **RPT1&2**, or **EXT**) for the volume envelope, the **HOLD** function can keep the repeats going without holding a key. See section 6 – *ARPEGGIATOR, DUO & HOLD* for details.

IMPORTANT: If the **TRIGGERING** switch is set to repeat or external modes, its position isn't affected by changing voices, so special triggering setups remain active. If you **STORE** a voice, the switch position is saved and will be recalled when that voice is selected—*provided* the switch is in **SG** or **MULT** at selection time.



External Triggering

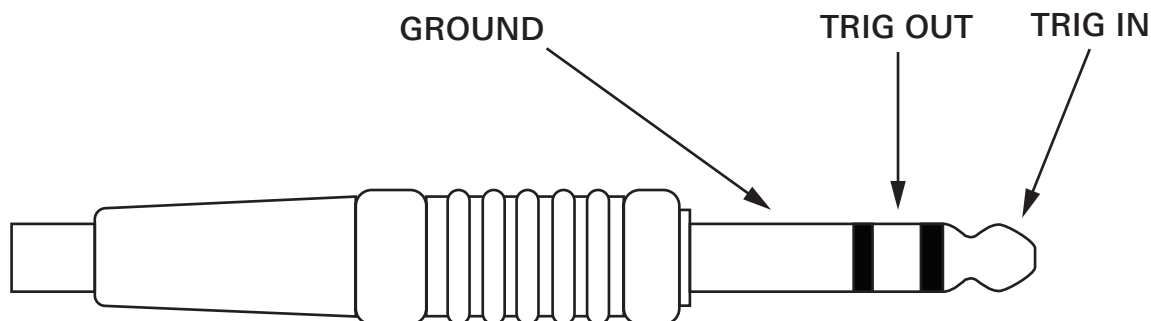
Connections and Signal Specifications

The **TRIGGER** jack socket provides separate connections for trigger in and out, using a *TRS* (Tip-Ring-Sleeve) connector.

TRIGGER OUT comes from the *RING* connection.

TRIGGER IN goes to the *TIP* connection.

The *RING* (**TRIGGER OUT**) is normally at 0 volts, but produces a +5V, 4 ms pulse each time the **VOLUME** envelope generator is triggered. During sequence playback, a pulse is sent on every sequence beat—useful for syncing other gear.



A trigger *IN* pulse to the **TIP** connection should last at least 4 ms, with at least 30 ms between the negative edge of one pulse and the positive edge of the next. Triggering happens on the *positive* edge. Reliable operation is possible at over 20 beats per second. ***The pulse must not exceed +5V.***

You can also trigger using *audio* signals - for example, sync pulses from a multi-track tape recorder or sounds like bass or snare drum hits. Try experimenting with different audio sources and levels for best results. You can even use the OSCar40's own sounds (via the envelope repeat function) to generate timing pulses and record them to an unused track on your multi-track tape or DAW.

The audio level entering the trigger *IN* connection should be at least 2 volts RMS and ***MUST NOT EXCEED*** +5v.

NOTE:

As **TRIGGER IN** is through the *TIP* connection, a normal *mono* jack plug can be used if trigger out is not required.

Using External Triggering IN

There are two methods of using trigger pulses from an external source:

1) DIRECT TRIGGER

If **TEMPO** is set fully anticlockwise to **EXT**, the **TEMPO CLOCK** is replaced by external trigger pulses, regardless of the **TRIGGERING** switch position. This is the easiest and recommended method, allowing all **TRIGGERING** switch functions to work with an external trigger.

2) DIVIDED TRIGGER

For **EXTERNAL TRIGGER DIVIDE**, set the **TRIGGERING** switch to **EXT**. All triggering will then come from a divided-down version of the incoming trigger pulses set by the **TEMPO** control.

In this mode, the **TEMPO** control divides into 8 zones, each corresponding to a different divide factor (1, 2, 3, 4, 6, 8, 12, 16). Set the knob near the centre of the desired zone.

To *reset* the divide function so the next external pulse produces a trigger, press and release the **SEQ** button. This works just like resetting the sequencer.

The divide function allows, for example, more efficient use of sequencer space when you want a sequence note to change only on every other beat of the basic timing signal.

Note that the divide function works differently when using MIDI clock, see section **9 - MIDI FUNCTIONS**.





5 VOICES

Control Summary

Switching On

When first powered on from the factory, the instrument automatically loads voice **1** from **ROM BANK A** (the original *OSCar* factory bank). If the *OSCar40* has been used before, it will load the last voice that was previously selected.

Selecting Voices

To *ensure* voices load and play immediately, set **TRIGGERING** to **SG** or **MULT** and **FUNCTION** to **NORM**. If you experience issues like no sound, check and restore these settings before troubleshooting further.

The Panel Voice

VOICE + key **0** selects the *PANEL VOICE* and sets up the following things:

OCTAVE to 8 ft (centre LED of display)

TUNE/TRANPOSE to **0** (A=440Hz)

INTERVAL to **0** (same semi-tone as **OSC1**)

FILTER DRIVE to 2/3 (start of thickened line)

ARPEGGIATOR to **UP/DOWN** and **HANDS-ON**

Oscillator waveforms to **WAVEFORM** switches settings (i.e. they will *not* be special waveforms)

Pressing and releasing the **VOICE** button selects the special *PANEL VOICE*. This is like selecting voice **0**, but with all parameters set to their last-used *PANEL VOICE* values.



The Scope of the Instrument

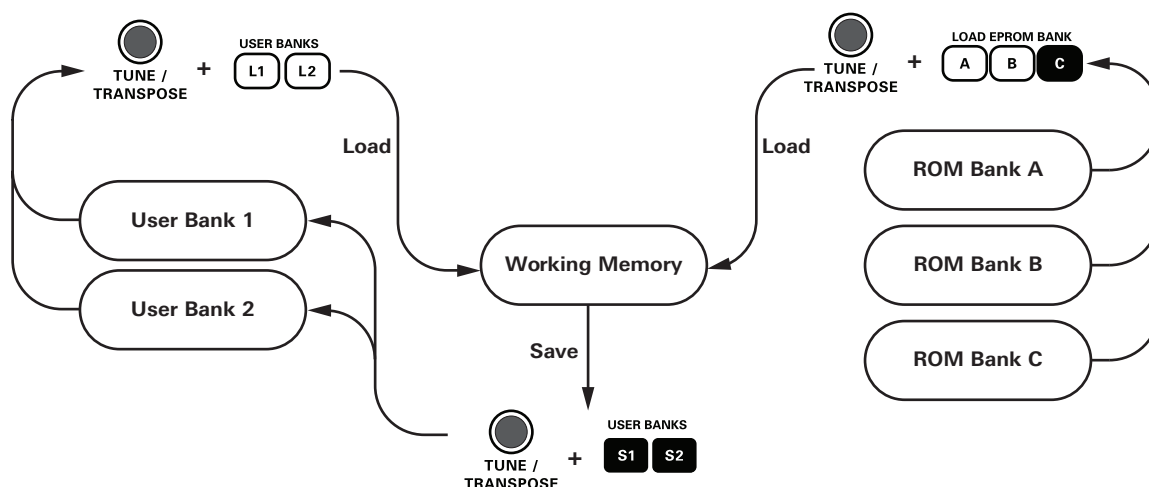
There are several ways to create different sounds with the instrument. The easiest is to use any of the 108 pre-programmed voices across Banks **A**, **B**, and **C** (36 in each). These voices can be set instantly, require no special experience, and offer a wide variety of sounds showcasing the instrument's potential.

Alternatively, you can select a voice and modify it, then store the result in one of the 36 voice locations. Once saved, it's as easy to select as any preset voice. You can continue to select, edit, and re-store voices—either in new locations or by overwriting existing ones—until you achieve the desired sound. The entire *bank* of 36 voices can be save to one of two battery backed up **RAM** locations.

You can also build your own voices from scratch. If you're new to synthesizers, it's easy to feel lost at first—but you're not alone. The best way to learn is to explore, practice, and get comfortable with the instrument's capabilities.

Whenever you store a sound in one of the 36 programmable voice locations, the exact setup is captured—whether it's a modified, original, or preset voice. Programmable voices can be recalled as often as needed and stay unchanged until reprogrammed. They're retained even after the instrument is turned off.

Understanding the OSCar40 Memory Structure



The OSCar40 features three ROM banks of 36 voices each, for a total of 108 pre-programmed voices. ROM Bank **A** is suspected to be the original OSCar factory bank. By default, ROM Bank **A** is loaded into the *working memory bank* at the factory and if the instrument is new, voice **1** from ROM Bank **A** will be loaded on first power up. The *working memory bank* is always the bank that voices are selected from.

There are also two RAM banks available for storing your own voices. From the factory these two *USER BANKS* are empty.

Working memory is retained after switching off the OSCar40, and will be reloaded when switched back on. However, loading a ROM bank or RAM bank will overwrite the *working memory bank*. If you have made edits to voices or created new ones and stored them to *working memory*, you have to save the *working memory bank* to one of the two RAM bank locations before loading new banks of voices, otherwise your work will be lost.

Loading and Storing Banks

To store the working memory to a *USER RAM* bank, press and hold the **TUNE / TRANSPOSE** button and then press the key labelled **S1** (-11) or **S2** (-9), then release the button and key to complete the store process. Be careful not to accidentally press **L1** or **L2**. It sucks when this happens.

USER RAM banks are loaded in a similar way, press and hold the **TUNE / TRANSPOSE** button while pressing the key labelled **L1** (-12) or **L2** (-10).

Moving Voices Between Banks

Changing banks doesn't immediately overwrite the current loaded voice. In this way you can move voices between user banks. You just have to remember where the empty voice locations are in the newly loaded bank. I keep a *stick-it* note on the synth, and note the last voice location used to save.



Selecting Voices

Each of the 37 keyboard keys is linked to a different voice memory locations. The numbers above the keyboard help identify them: keys **-12** to **-1** and **1** to **24** are for programmable voices, while key **0** selects the *PANEL VOICE*.

To load a voice, hold the **VOICE** button (in the **STORE AND RECALL** section), press your chosen key while holding **VOICE**, then release both. The selected voice is now ready to play.

VOICE + key **-12** to **24** then release
loads one of the 36 voices from the currently loaded bank (ROM banks **A**, **B** and **C** or User banks **1** and **2**).

The **TRIGGERING** and **FUNCTION** switches are only saved with a selected voice if they're physically set to their programmable positions:

TRIGGERING positioned at **SG** (single) or **MULT** (multiple)
FUNCTION positioned at **NORM** (normal)

If the **TRIGGERING** or **FUNCTION** switches are not in *programmable* positions, their current knob settings override the saved voice settings. This lets you maintain special setups - like arpeggiation with envelope repeats - when changing voices.

Storing Voices

Storing a voice for future use is similar to selecting one, but you'll use the **STORE** button as well:

1. Hold down both the **VOICE** and **STORE** buttons.
2. While holding them, press a key from **-12** to **-1** or **1** to **24**.
3. The current sound will be saved to that key number and can later be recalled by pressing the same key with the **VOICE** button alone, as explained in Selecting Voices.

TRIGGERING and **FUNCTION** settings are always stored, but to recall these settings accurately, make sure the switches are set to their *programmable* positions when selecting the voice.

VOICE + **STORE** + key (**-12** to **24**) stores all (effective) control settings (except **FINE TUNE** and **TEMPO**) in that voice location. The previously stored voice in this location will be overwritten.

VOICE + **STORE** + key **0** saves the edited voice back to its *original* location. This is a convenient short cut for saving edited voices without having to remember which location it originally came from.

The Panel Voice

The *PANEL VOICE* (voice **0**) is selected just like a programmable voice, but with key **0**. When *PANEL VOICE* is selected, certain settings are automatically initialized:

OCTAVE to 8 ft (centre LED of display)

TUNE/TRANPOSE to **0** (A=440Hz)

INTERVAL to **0** (same semi-tone as **OSC1**)

FILTER DRIVE to 2/3 (start of thickened line)

ARPEGGIATOR to **UP/DOWN** and **HANDS-ON**

Oscillator waveforms to **WAVEFORM** switches settings (i.e. they will *not* be special waveforms)

This automatic reset provides a good starting point for building new sounds. If you're unsure what this means, just ignore it for now and explore more as you become familiar with the instrument.

Selecting the *PANEL VOICE* and playing keys may not produce any sound if the controls aren't set properly - don't forget to turn up the volume. You may need to adjust waveforms and envelopes to hear anything, or to avoid unpleasant sounds.

Once you've set up the controls as you like, you can use voice **0** as a 37th *programmable* voice, but automatic resets like centre **OCTAVE** and zero transpose will always apply. To avoid this, press and release the **VOICE** button without pressing any keys. This recalls the **PANEL** voice and keeps the octave, transpose, arpeggiator mode, and waveforms as they were when the *PANEL VOICE* was last used.

The word (**PANEL**) below the voice button reminds you of this secondary function. If this isn't clear now, you can ignore it and revisit the function later as needed.



Editing Voices

When you load a voice, the physical control positions will usually differ from those stored in the voice. The **OCTAVE** display blinks about once per second to indicate that some panel controls aren't set correctly for the loaded voice.

Unfortunately, there's no visual indication showing the optimal control positions for the selected voice. (Ideally, the controls would automatically adjust to the correct settings for each voice—though this is difficult to implement, unless you're *Melbourne Instruments* :P)

Moving any control will instantly edit the sound from its saved state. To locate a control's saved position, use the **EDIT FIND** function.

The EDIT FIND Display

Hold the **STORE** button and watch the **OCTAVE** display while turning a control. If LEDs on the left light up, turn the control clockwise; if LEDs on the right, turn it anticlockwise. As you approach the correct position, more LEDs will light up. When all are lit, you're very close; a few small adjustments will make the display disappear, and the control will be active.

If the **FIND** display did not appear at all, the knob must have been in the correct position to start with.

EDIT FIND might seem tricky at first, but once you get used to it, you can quickly locate the correct control position. This helps you see how a voice was originally set up before making changes, and allows for smooth adjustments - avoiding abrupt sound changes from sudden setting jumps.

This is referred to elsewhere as **EDIT FIND AND RELEASE**. Once you've used this function for a control, the **FIND** display won't work for that control again until you select a new voice. Selecting a voice re-inhibits the controls (except those already matching the saved position) and discards any 'edited' settings.

Storing the Newly Formed Sound

You can store the edited sound at any time in a programmable voice slot using the process outlined in **STORING VOICES**. To save it back to its *original* location, press and hold **STORE**, then press key **0**.

Be sure to store your sound if you want to use it later - selecting a new voice or switching off the instrument will erase any unsaved edits. Also, storing a voice in a location (-12 to -1 or 1 to 24) will overwrite the previous voice saved there.

IMPORTANT: When storing edited voices, don't hesitate on the voice button before pressing your chosen programmable key, or you could accidentally select the *PANEL VOICE* instead. Selecting the *PANEL* or *(PANEL)* voice (noted by brackets) releases all control settings, making the entire panel define the sound. If this happens, any part-edited sound will be lost.

Creating Original Voices

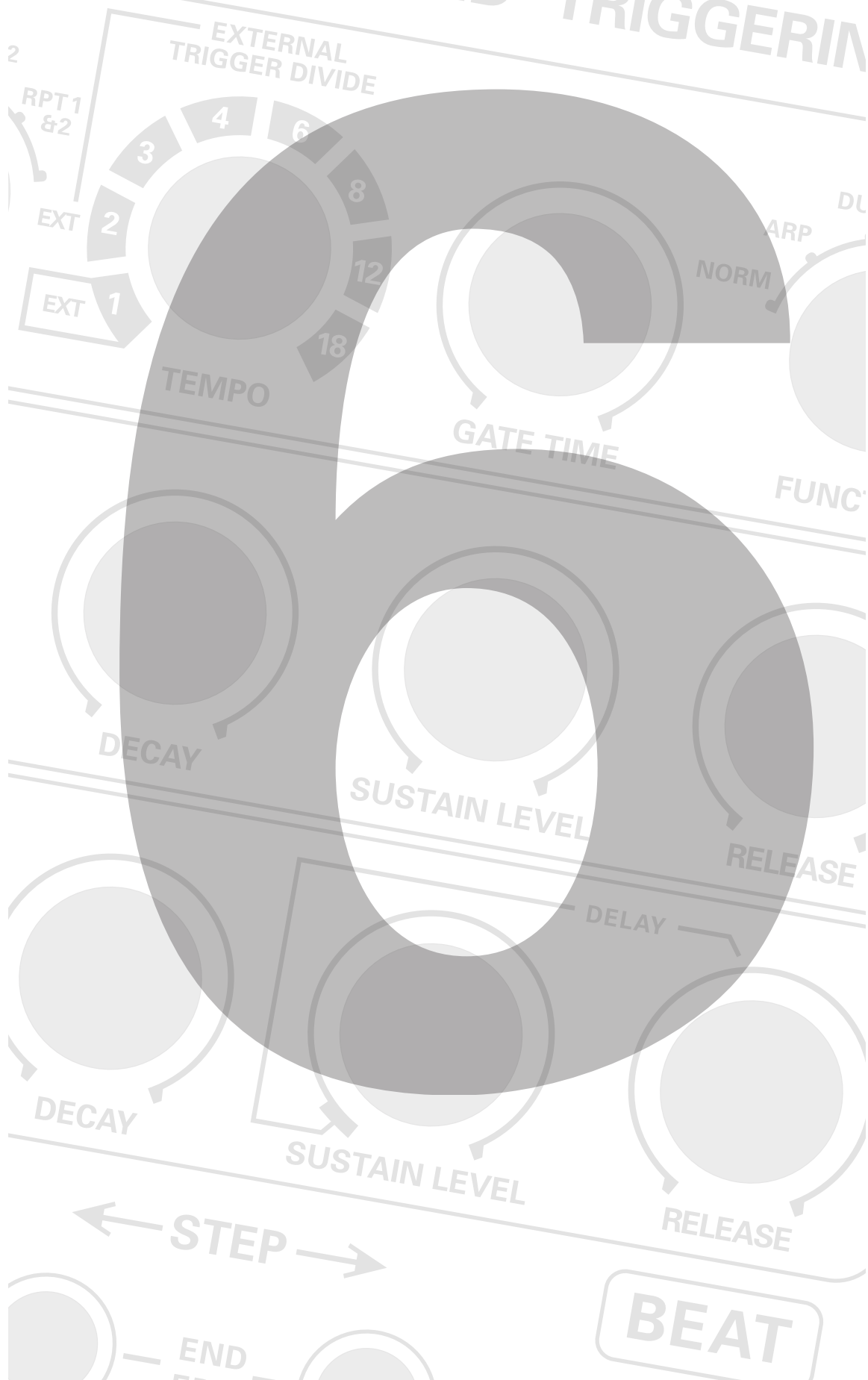
Creating voices from scratch means not using the ROM bank presets. Simply select the *PANEL* voice (voice **0**) and start building your sound. The special *(PANEL)* voice can also be used, but it's less ideal since settings like transpose, arpeggiator mode, and waveform might not be obvious.

Using the instrument in this way is like using a non-programmable synthesizer, and it requires some understanding of all the voice-forming controls. This approach unlocks the instrument's full potential, but may take practice to master. All the typical controls expected in a non-programmable synth are provided.

Advanced features, like waveform building, are optional and not essential for starting from scratch. Other sections of the manual cover every function in detail. While the manual may not suffice for absolute beginners, studying it thoroughly is helpful before seeking a synthesizer book or expert advice.



... AND TRIGGERING



6 THE ARPEGGIATOR, DUO AND HOLD

Control Summary

The FUNCTION switch - Manual and Programmable Operation

Although not directly related, the arpeggiator, duo, and hold features are covered together here because they're all activated via the **FUNCTION** switch. Before delving into these three functions, this section explains how to use the switch.

When the switch is set to **NORM** (fully anticlockwise), its function is programmable, meaning the selected voice sets the function based on how it was stored. In *any* other position, the function is determined solely by the switch, regardless of the setting in voice memory. This is similar to the **TRIGGERING** switch, which is only programmable in **SG** and **MULT**.

This setup lets you keep any of the five function combinations—arpeggiator or duo, with or without hold, or hold alone - even when switching to voices with different **FUNCTION** settings (such as the ROM voices, which default to **NORM**).

When storing a voice, the switch's effective function is saved with it. Loading that voice later (with **FUNCTION** in **NORM**) immediately restores the stored function. If you don't want arpeggiator, duo, or hold active by default, be sure to switch **FUNCTION** back to **NORM** before storing.



Hold

The three most clockwise positions of the **FUNCTION** switch give the same functions as the other three positions except that the *gate* (described in section 4 - **TRIGGER, GATE AND ENVELOPES**) is kept on. It has the same effect as holding keys down on the keyboard indefinitely.

The three most clockwise **FUNCTION** switch positions provide the same functions as the preceding three, but keep the gate *on* - just like holding keys down indefinitely on the keyboard. (See section 4: **TRIGGER, GATE AND ENVELOPES** for more details.)

The Arpeggiator

Set the **FUNCTION** switch to **ARP** or **A-HOLD** for indefinite hold. Every held key will now be played in ascending or descending order, depending on the selected *arpeggiator mode*.

If needed, you can memorize the notes played so the arpeggiation continues without holding down the keys. Four **SEQUENCE EDITING** buttons have extra functions to set the operating mode, but they only take effect when the arpeggiator is *running*.

Speed Adjustment and Triggering

The **TEMPO** control sets the arpeggiator speed, just like it does for sequencer playback and envelope repeats. If external triggering is used, arpeggiator speed will follow the external trigger pulses.

While running, the arpeggiator doesn't trigger the envelope generators with each note change. You can use the envelope repeat settings on the **TRIGGERING** switch to automatically trigger either or both envelope generators for every new note. See section 4 – **TRIGGER, GATE & ENVELOPES** for details on these options.

Setting Direction of Play

Direction is set by the two **STEP** buttons in the **SEQUENCE EDITING** section:

← STEP (then release)
sets repeated descending play to ***DOWN MODE***

STEP → (then release)
sets repeated ascending play to ***UP MODE***

← STEP + STEP → (then release) (both step buttons)
sets alternate ascending/descending play ***UP/DOWN MODE***

TUNE/TRANSPOSE + key ARP RND / ON (8) (then release)
sets the play back to random. Notes may be repeated.

When random ARP mode is enabled, order of notes is constantly updated and notes may repeat before all other notes have been played. Notes may also play twice in a row.

When setting the direction of play (*excluding random mode*), remember that the **STEP** buttons are also used for starting and stopping sequence playback. To prevent the sequencer from starting, hold down at least one key of the keyboard while the arpeggiator direction is being set.



Hands-On Mode or Memory Mode

This is set using the **INSERT** and **DELETE** buttons in the **SEQUENCE EDITING** section:

Pressing and releasing **DELETE** sets the arpeggiator to **HANDS-ON** mode. In this mode, only the keys currently held are included in the arpeggiation - release a key and it's dropped from the pattern. If all keys are released, the oscillators lock to the last played note (audibility depends on envelope and **HOLD** settings). **HANDS-ON** is ideal for live playing, as it responds instantly to keyboard changes.

Pressing and releasing **INSERT** sets the arpeggiator to **MEMORY** mode. In this mode, you can keep adding notes to the arpeggiator memory as long as you hold *at least one key* or the **SPACE** button.

Once *all* keys (or the space button) are released, pressing a key or the **SPACE** button again will clear the memory and let you enter a new set of notes.

This mode works best with **HOLD** active (or a long envelope release), letting the instrument play unattended while you use another. When new notes are needed, just play them to instantly clear and replace the previous set.

TUNE and **INTERVAL** transpose controls can be used without clearing the arpeggiator memory - just hold the **TRANPOSE** button before pressing a key in the transpose range.

Arpeggiation in Stored Voices

When a voice is stored, the current arpeggiator **MODE** setting will be stored as well as the position of the **FUNCTION** switch.

Note: the **PANEL** voice and all the **ROM** voices initially set the arpeggiator modes to **UP/DOWN** and **HANDS-ON**.

The Duo Function

Duophonic Keyboard Playing

Set the **FUNCTION** select switch to **DUO** or **D-HOLD** (if indefinite hold is required). Now, if two keys are played, both notes will sound (one oscillator each).

If *all* keys are released, pressing a single key will force *both* oscillators to the same note. This allows you to easily switch between unison and harmony using just the keyboard.

DUO doesn't provide true duophonic performance, but it does expand the instrument's capabilities by allowing keyboard-controlled harmony - something rare for monophonic synths. However, keep the following points in mind:

It can be tricky to know which oscillator is playing which note in **DUO** mode. The rule is: after releasing all keys, the first key you press will lock oscillator 1 to that note until you let go of that key.

Polyphonic synthesisers assign voice blocks to keys, usually with the same sound - otherwise, it's hard to track which sound is on which key. For best **DUO** results, set both oscillators to the same waveform, with **OCTAVE SHIFT** and **INTERVAL** at **0**, and **DETUNE** close to **0**. This ensures both oscillators sound alike.

FILTER TRACKING, if active, is tied to oscillator **1**. With both oscillators using the same filter, critical filter settings might cause inconsistent results—unless you play very carefully. For a more reliable **DUO** response, use **NO TRACK** filtering.



← STEP



— END
EDIT —



NGLE/STOP

START/REPEAT

EDITING

ARP RND

CC DU

ON

OFF

SEQUENCES

6

7

8

9

7 THE SEQUENCER

Introduction

The sequencer is a comprehensive section for storing notes, spaces, chains of sequences and voice changes which can later be played automatically.

The sequencer can hold up to 22 independent sequences at once, sharing 1,500 event spaces. While 1,500 events may sound limited if used for just single notes, the effective capacity increases with chained repeated phrases - allowing for thousands of notes in practice.

The first 12 sequences (**SEQUENCES 1–12**) handle only notes, spaces, and repeats, making them ideal for simple patterns.

The next 10 sequences (**CHAINS 15–24**) can be used like sequences **1–12**, but can also include voice changes and *references* to sequences **1–12**, plus notes and spaces, in *any* order or *combination*.

Notes within sequences can be *extended* (tied) over any number of beats such that no envelope trigger occurs between them. Similarly notes can be carried over to the *next* note with no triggers in between (legato phrasing).

To create sequences, enter the **SEQUENCE EDITING** mode. Here you can freely create, view, or edit the selected sequence without risking changes to any others.

In editing mode, you can delete, replace, or insert events until you reach the maximum of 1500 events for a sequence or the total storage limit. If you run out of space, make room by shortening or clearing another sequence.

Remember: you can save sequences with a **SYSEX** dump, so always save before clearing to avoid data loss. (see section **9 - MIDI FUNCTIONS**)



While editing, you can play any of the simple sequences (1–12) for reference without leaving editing mode. The **OCTAVE** display reverses (LED off indicates octave) to remind you you're editing. *Note:* the **STORE** button works differently here—sequences are stored as you edit them, so there's no need for a separate save step after editing.

- Created sequences remain saved in the sequencer's memory even when powered off.
- Sequences must be edited individually if changes are needed.
- Any sequence can be instantly selected for playback.
- Playback speed can be set by the internal **TEMPO** clock or by external trigger pulses (e.g., from a drum machine).
- External trigger pulses can be divided by various factors:
- Synchronize with an external source if its rate is a multiple of the desired sequencer rate.
- Enable polyrhythmic effects by triggering sequences on irregular pulse counts.
- For more on polyrhythms and triggering: see section 4 - *TRIGGER, GATE AND ENVELOPES*.

While a sequence is playing, the *GATE TIME* control allows the *on* period (simulated *key down* time) of each note to be adjusted. (This is in addition to the normal envelope controls.)

Getting Acquainted with the Sequencer

There are quite a few sequencer commands to learn, which may seem complicated at first. But don't be discouraged. If you feel overwhelmed by the number of commands, start by working through the sequencer section in section 3 - *STARTING TO PLAY* to get comfortable.

If things aren't working as expected, the best approach is to start fresh. Switch to editing mode, clear the sequence, add a few simple notes and rests, finish editing, and play it back. Focus on creating basic sequences first—don't worry about advanced features like step-through, inserting or deleting notes, or repeat commands until you're more comfortable. Follow the command instructions closely, as they do exactly what is described—nothing more, nothing less. Remember, you don't need to use every feature to create complex sequences!

Creating and Editing Sequencers

When powered on, the sequencer enters **STANDBY** mode, ready to play sequence **1**. Before creating or editing a sequence, set up the following:

- Set the **FUNCTION** switch to **NORM** (other settings may give unexpected results)
- Set **TEMPO** to approximately the 2 o'clock position (for a fairly fast tempo)
- Set **GATE TIME** to about the 2 o'clock position (for a fairly short gate time)

ENTERING THE SEQUENCE EDITING MODE

SEQ + STORE + key 1 to 12 puts one of the simple sequences into the **EDITING** mode.

OR

SEQ + STORE + key 15 to 24 puts one of the **CHAIN** type of sequences into the **EDITING** mode.

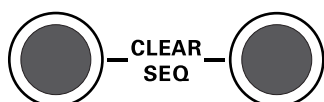
At this point, you can either add events to the end of the existing sequence or start entering events into an empty sequence.

You now have access to the following sequence editing commands:

CLEARING THE SELECTED SEQUENCE

INSERT

DELETE



INSERT + DELETE (CLEAR SEQ) together **ERASES THE ENTIRE SEQUENCE** - be careful !!

Note: It's best to press the **INSERT** button first; otherwise, the delete function will activate and play the deleted event. While this isn't usually a problem, the deleted event could be something disruptive, like a voice change.

You can only clear a sequence while in editing mode, and only the selected sequence can be cleared.

- There is no command to clear all sequences at once for safety reasons.
- Each sequence must be entered into editing mode and cleared individually.



Entering Events into the Sequence

An **event** refers to anything that can be stored in a sequence.

- **Simple sequences (1–12)**: Can store notes (normal, tied, or legato), spaces, and repeat commands.
- **Chain sequences (15–24)**: Can store all of the above, plus voice changes and references to any of the simple sequences.

Think of chain sequences as songs or arrangements, since they can combine multiple sequences and events.

Like clearing a sequence, the following of functions are only possible in the **editing mode** and only operate on the selected sequence.

1) ENTRY COMMANDS AVAILABLE IN ANY SEQUENCE

Playing a key when **no other keys** are down enters a normal note into the sequence. The note plays normally as it enters the sequence.

Playing a key **before** releasing the previous key enters a **LEGATO** note into the sequence. The note plays legato style as it enters the sequence. Any number of notes can be run together this way.

STEP →

while holding down the previous key extends the previous note for another beat (adds a **TIED** note). No change is heard in the note but the **BEAT** LED flashes to confirm the entry. The note can be extended any number of ties but the **repeat event** command can be used to save space if a very long tied note is required.

SPACE

when no other keys are down enters a space (rest period) into the sequence. In the editing mode, a burst of **NOISE** is heard to help in identifying spaces during subsequent editing. During playback, it produces the proper effect, just a space.

REPEAT EVENT + key 1 to 24

enters a **repeat previous event** command into the sequence. The number of repeats makes the **TOTAL** up to the key number selected (e.g. key **4** gives 3 repeats plus the previous note). In the editing mode, the number of repeats (key number less one) are sounded out as beats of **TOP C**. During playback, it produces the actual note. **ANY** event can be repeated using this command.

2) ENTRY COMMANDS ONLY AVAILABLE IN CHAIN SEQUENCES

VOICE + key -12 to 24

enters the selected voice into the sequence including the *PANEL VOICE* if required. The special (*PANEL*) can also be entered by pressing **VOICE** then releasing it. In the edit mode, as well as the voice change taking place as it enters the sequence, a *dummy* rest period and a flash of the **BEAT** LED occur. This helps in identifying voice changes during subsequent editing. Any number of voice changes can be entered anywhere in the chain.

SEQ + key 1 to 12

enters one of the *SIMPLE* sequences into the *CHAIN* sequence. The simple sequence is played out immediately it enters the chain.

3) STEP AND JUMP COMMANDS - (*available any time during editing*)

The step and jump commands are for moving around in a sequence before changing, inserting or deleting events or for simply checking the sequence. They are not entry commands and do not affect the sequence itself.

← STEP then release

steps back one event in the sequence and plays it in the same way that it played when it was being entered. When the beginning of the sequence is reached, it has no effect.

STEP → then release

steps forward one event in the sequence and plays it as with the step back command. Note that when stepping forward through a sequence *FROM THE BEGINNING*, the first step plays the *SECOND* event in the sequence; it is not possible to hear the first event using this command (use the *step back* command to obtain the first event). When the end of the sequence is reached, it is possible to make *ONE MORE FORWARD STEP* beyond the last event in the sequence. Nothing is played but the **BEAT** lamp will flash. If more events are to be added to the sequence, it is important to make this final step or the last event will be *REPLACED* by the next entry command.

REPEATED STEPPING

To step repeatedly, do not release the step button. The initial press steps through one event as above but after *missing a beat* (to simplify single stepping) the sequence begins to play at the normal rate as long as either **STEP** button is held down. Adjust **TEMPO** to give the required playing speed. This is useful for stepping quickly through a long sequence.



SEQ



(RESET)

SEQ (RESET) then release

causes a **JUMP** to the **START** of the sequence. Nothing is played but subsequent use of the step commands will reveal that the jump has been made. It is useful for rapidly moving to the beginning of a long sequence without having to play through it. (This command can also be used in sequence playback for cueing sequence and drum machine start-ups.)

REPEAT EVENT



(JUMP TO END)

REPEAT EVENT then release (marked as **JUMP TO END**)

causes a **JUMP** to the **END** of the sequence. Like the *jump to start* command nothing is played and its purpose is to move rapidly to the end of the sequence, after stepping through and checking etc. After using this command, the situation is just the same as after entering the editing mode - its ready to add more events to the end of the sequence.

4) DELETE, INSERT AND REPLACE FUNCTIONS

- (available any time during editing)

INSERT

DELETE



DELETE

removes the event that has just been entered or played out using the step commands. The deleted event plays to confirm the operation. Unless the deleted event is the last in the sequence, all the events that come after the deleted one are shifted back one beat to *fill in* the gap.

INSERT

DELETE



INSERT + any valid *ENTRY COMMAND* in **1)** and **2)** above

adds the event into the sequence immediately *BEFORE* the event just played using the step commands. If it is used during normal entries (i.e. at the *END* of the sequence), it makes no difference to the entry command. The event plays out or takes effect in the normal way as it enters the sequence. All events in the sequence that comes after the inserted one are shifted forward one beat.

Note that voice changes and simple sequences can be *INSERTED* into chains. For example to insert a voice change, hold down **INSERT** and **VOICE** and then press the appropriate voice key.

REPLACING EVENTS

While stepping through a sequence, you can replace the event that was just played by using a new *ENTRY* command—this simply overwrites the existing event without changing the total number of events in the sequence. This is different from *inserting*, which *adds* a new event to the sequence and increases its length.

Remember: After checking or modifying a sequence, step forward one step beyond the last event before entering new events. If you don't, your next entry will overwrite the last event. A dummy LED beat will confirm you're past the end. Alternatively, you can safely use the *JUMP TO END* command (press and release **REPEAT EVENT**) to position yourself for new entries.

If the sequencer runs out of space the OSCar40 will simply stop entering new events. It will appear to have stopped responding.

While sequences are being created or edited, remember that the sequence space is limited in two ways:-

- Each individual sequence can hold up to 255 events.
- The combined total for all sequences and chains is 1500 events.

If you reach either limit, you'll get *no response* when trying to enter another event. To continue, shorten or clear another sequence to make room.

If you need a very long sequence without repeats, split it into 2 or 3 parts.

Use the chaining feature to link these parts, which will only use a few events for the chains.

If parts of a sequence are repeated, you can save space by using the *REPEAT EVENT* command or by chaining together simple sequences. It's generally easier and more efficient to break the sequence into short, simple phrases and then chain them as needed. This approach makes the best use of available space and rewards a bit of advance planning.



5) CHECKING EXISTING SIMPLE SEQUENCES WITHOUT ENTERING THEM - *(available at any time)*

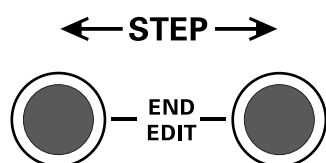
If you're unsure what sequences are stored in group **1–12**, you can play them back without interrupting the current editing session. Just use the **MINUS** keys along with the sequence button to do so.

SEQ + key **-1** to **-12**

This function plays back the selected sequence without entering edit mode. You can adjust **TEMPO** for the desired speed. The sequence being edited is retained and you can play back any sequence - simple or chain - during editing. When editing a simple sequence (1–12), you can use it to test that sequence directly without leaving editing mode.

ENDING THE EDIT

When you finish editing a sequence, you must exit editing mode to use the instrument normally. Alternatively, you can start editing a different sequence. Your options are:



SINGLE/STOP **START/REPEAT**

1) PRESS THE 2 STEP BUTTONS SIMULTANEOUSLY (marked **END EDIT**)

This puts the sequencer into the **STANDBY** mode leaving it ready to play the sequence just edited (or for normal keyboard playing).

2) SELECT ANOTHER SEQUENCE READY TO PLAY (i.e. **SEQ** + key)

The only difference between this and the first method is that the sequencer will be ready to play the newly selected sequence instead of the one that has just been edited.

2) START EDITING ANOTHER SEQUENCE (i.e. **SEQ** + **STORE** + key)

This has exactly the same effect as starting an edit from the **STANDBY** mode. The **OCTAVE** display will blink OFF to confirm the operation.

Playing Sequences

When power is first applied, the sequencer is in the **STANDBY** mode and sequence **1** is ready to play, **STANDBY** merely signifies that the selected sequence is **READY FOR PLAYBACK**. The sequencer is always in this mode when it is not playing or editing.

Selecting a Sequence Ready for Playback

SEQ + key 1 to 12

selects a simple sequence ready for playback but does not actually start it playing.

SEQ + key 15 to 24

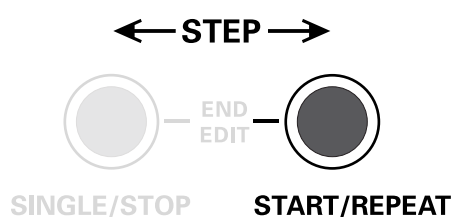
selects a **CHAIN** type of sequence in the same way.

In either case:-

If a sequence is playing it will stop. The **OCTAVE** display will blink all *on* if the sequence selected is a *different* one from the currently selected sequence.

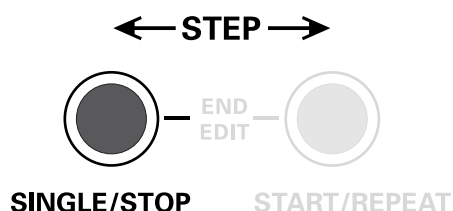


Starting and Stopping Playback



STEP → (START/REPEAT)

- Pressing once will playback the sequence *once*.
- Pressing *twice* in quick succession will put sequence playback into *repeat mode*.
- If the sequence is already playing, pressing again will put sequence playback into *repeat mode*.



← STEP (SINGLE/STOP)

- Pressing once will cancel *repeat mode* and continue playback until the *end* of the sequence.
- Pressing *twice* in quick succession will stop playback *immediately*.
- If the sequence is in *single play* mode, pressing once will stop playback immediately.

SEQ



(RESET)

SEQ (RESET) then release

unconditionally **RESETS** the sequence to the start **WITHOUT AFFECTING THE RUN OR REPEAT STATUS**.

This function helps ensure that sequences start correctly when triggered by an external device. For example, if you're using a drum machine and a sequencer that follows its tempo, stopping the drum machine can cause the sequencer to halt partway through its sequence. By performing a sequencer **(RESET)**, you can make sure the sequencer returns to the start.

Then, when you restart the drum machine at the beginning of its rhythm pattern, the sequencer will also start from the beginning, keeping everything synchronized.

Note that if the *EXTERNAL TRIGGER DIVIDE* is being used, the *RESET* command sets up the divider so that it will produce an internal trigger on the *NEXT* incoming trigger pulse.

Immediate Playback of the Simple Sequences

Press **SEQ** with any of the **-1** to **-12** keys to instantly play simple sequences **1-12**, just like previewing in edit mode. This doesn't affect the sequencer's status and works at any time, even during playback.

SEQ + key -1 to -12

Instantly plays one of the simple sequences without using the normal **START/STOP** buttons. This operation does not affect the sequencer's status.

If an *immediate* play is already running, you can set up one to follow it after the first finishes. This process can continue indefinitely, but only one additional *immediate* play can be queued while another is active.

If a sequence is playing in the *normal* way, it will interrupt it immediately. When the *immediate* play has finished, the *normal* playback will be resumed.

Note that if the *normal* playback is a chain and a *chained* simple sequence is currently running, the *immediate* play that has been requested will only start when the *chained* simple sequence has finished.



Sequencer Summary

Editing

ENTERING THE EDITING MODE

SEQ + STORE + key 1 to 12 (SEQUENCES) OR 15 to 24 (CHAINS) puts the selected sequence into *EDITING* mode at the *END* of the existing sequence. If the sequence is empty then editing begins at the start of the sequence.

CLEARING THE SELECTED SEQUENCE

INSERT + DELETE together clears all data from the selected sequence. (make sure to push the insert button down first)

ENTRY COMMANDS (ANY SEQUENCE)

PLAYING A KEY when no others are down enters a *NORMAL NOTE*

PLAYING A KEY before releasing the previous key enters a *LEGATO NOTE*

STEP → while holding down a key enters a *TIED NOTE*

SPACE when no other keys are down enters a **REST** (and plays noise to indicate the space has been added)

REPEAT EVENT + key 1 to 24 enters a *REPEAT PREVIOUS EVENT* by key number -1 times (plays a *TOP C* to indicate repeats)

ENTRY COMMANDS (CHAIN SEQUENCES ONLY)

VOICE + key -12 to 24 enters a **VOICE CHANGE** (a noise plays to indicate that it has been added)

SEQ + key 1 to 12 enters a *SIMPLE SEQUENCE* into the chain

STEP AND JUMP COMMANDS (NOT ENTERED INTO THE SEQUENCE)

← STEP (then release) steps back and plays one event

STEP → (then release) steps forward and plays one event

(Keep either button down for repeated stepping at **TEMPO** rate)

SEQ (RESET) then release causes a *JUMP* to the *START* of the sequence

REPEAT EVENT (JUMP TO END) then release causes a *JUMP* to the *END* of the sequence ready for more events to be entered

EDIT MODE

DELETE, INSERT AND REPLACE

DELETE removes the event that has just been entered or played

INSERT + ENTRY COMMAND adds the event *BEFORE* the one just played using the step commands

REPLACING EVENTS the event just played during stepping is *REPLACED* by the next *ENTRY* command

CHECKING SIMPLE SEQUENCES WITHOUT ENTERING THEM

SEQ + key 1 to -12 plays back the sequence without entering it and without changing edit status

ENDING THE EDIT

Press the 2 **STEP** buttons simultaneously (marked **END EDIT**) *or* load a different sequence to play (i.e. **SEQ + key**) *or* start editing another sequence (i.e. **SEQ + STORE + key**)

Playback

NORMAL PLAYBACK FROM THE STANDBY MODE

SEQ + key 1 to 12 or 15 to 24 selects a sequence in the *STANDBY* mode ready for playback

STEP → (START/REPEAT) starts a *SINGLE* play or sets *REPEAT* if sequence is already playing (a quick double press will start *REPEAT* play)

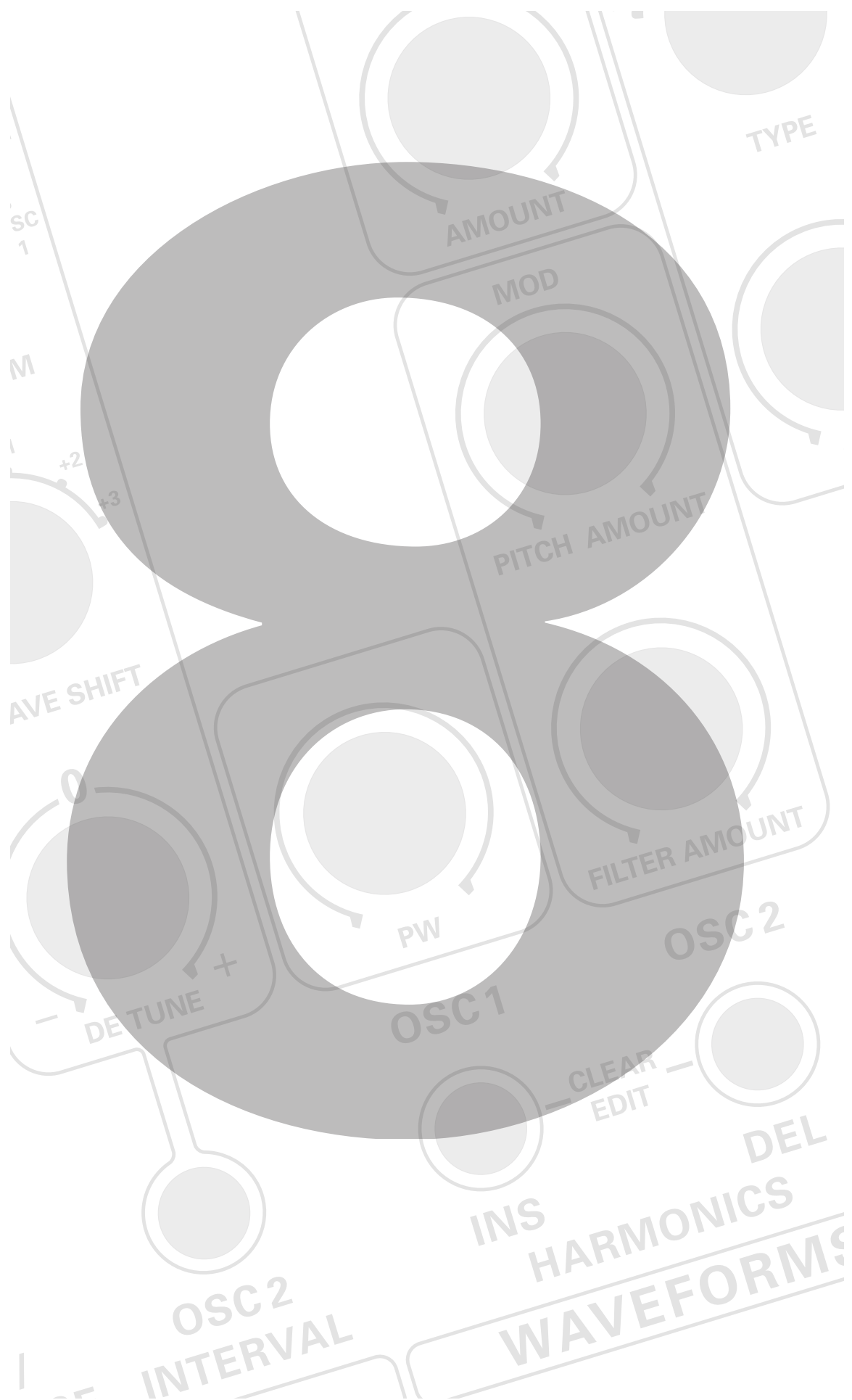
← STEP (SINGLE/STOP) cancels *REPEAT* if operation otherwise *STOPS* playback immediately (a quick double press will *STOP* play)

SEQ (RESET) then release *RESETS* the sequence to the start *WITHOUT AFFECTING THE PLAYBACK STATUS*.

IMMEDIATE PLAYBACK OF THE SIMPLE SEQUENCES

SEQ + key -1 to -12 plays one of the simple sequences immediately *WITHOUT AFFECTING PLAYBACK STATUS*. Subsequent *immediate* plays can follow on directly



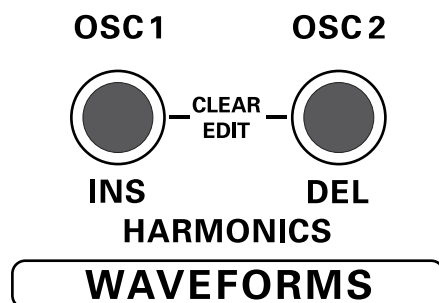


8 SPECIAL WAVEFORMS

What Waveforms are Available

Standard waveforms can be selected independently for the two oscillators using the **WAVEFORM** switches in the **OSC1** and **OSC2** sections of the panel. If oscillator 2 select switch is set to **OSC1**, then oscillator 1 select switch will control both oscillators waveforms.

This is described in section 3 - *The Control Panel*.



In addition to standard waveforms, users can choose from five special *preset* or five *programmable* waveforms, created by combining fundamental and harmonic sine waves. Each oscillator's waveform selection is entirely independent. For example, oscillator 1 might use a standard sawtooth waveform, while oscillator 2 uses or edits a programmable waveform. Any combination of waveforms is allowed, with no restrictions.



Selecting Special Preset Waveforms

OSC1 + key -3 to -7

selects one of the *5 preset* waveforms for oscillator 1

OSC2 + key -3 to -7

selects one of the *5 preset* waveforms for oscillator 2

To save time, if the *same* preset waveform is required for both oscillators, the two waveform buttons can be held down at the same time while the appropriate waveform key is pressed:

OSC1 + **OSC2** + key -3 to -7

selects the same preset waveform for both oscillators.

(Also note that if oscillator 2 **WAVEFORM** select switch is in the **OSC1** position, it is only necessary to select a special waveform for **OSC1**. The oscillator 2 waveform will follow automatically.

The 5 preset waveforms are:-

-3 ***FULL ORGAN***

containing square wave signals in lower octaves.

-4 ***HARPSICHORD***

containing sawtooth plus additional high frequency components.

-5 ***STRONG LEAD***

a complex pulse waveform with a powerful percussive sound.

-6 **DOUBLE PULSE**

waveform with a slightly *ringing* sound.

-7 ***RANDOM***

contains very high levels of high harmonics. It is most effective in the lower octave registers where these higher harmonics are in the audio band.

When these waveforms are selected, they will appear virtually instantly although the time taken for the ***FULL ORGAN*** waveform may be just noticeable.

Selecting Programmable Waveforms

The procedure is exactly the same as for selecting the special preset waveforms.

OSC1 + key -8 to -12

selects one of the 5 *programmable* waveforms for oscillator 1

OSC2 + key -8 to -12

selects one of the 5 *programmable* waveforms for oscillator 2

All the rules for the *preset* waveforms apply. The two oscillators can be set to different waveforms or they can be set simultaneously to the same waveform by holding down both waveform buttons etc. However, the programmable waveforms take some time to develop. This is explained further on in **CREATING WAVEFORMS**.

Even if the *programmable* waveform locations have not been programmed, there will probably be something interesting there.
Try them out!

Getting Back to the Standard Waveforms

To return to the standard waveforms, simply turn the appropriate **WAVEFORM** switch to the desired standard waveform. Disturbing the switch makes it release the special waveform. (See **EDITING VOICES** in section 5 - **VOICES**).



Creating Waveforms

First, it's advisable to enter **WAVEFORM EDITING** mode so you can hear the waveform as you create it. While this isn't essential—and there are benefits to not using the edit mode (see **RAPID WAVEFORM CREATION**) - being able to listen as you work can be helpful.

For best results, it's recommended (though not essential) to either put both oscillators in edit mode or turn down the one you're not editing using the **OSCILLATOR BALANCE** control. The suggested approach is to use both oscillators, detuning them slightly, with a long volume release time. Select the bottom octave, set the low-pass filter almost fully open, and keep the **Q** fairly low. Alternatively, to get something going quickly, select voice **13** (or **12**) from ROM Bank **A** and set the **FUNCTION** switch to **N - HOLD**.

Entering the Waveform Editing Mode

OSC1 + key **0**
puts oscillator 1 into the **WAVEFORM EDITING** mode.

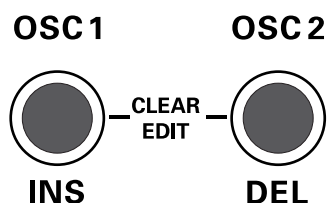
OSC2 + key **0**
puts oscillator 2 into the **WAVEFORM EDITING** mode.

This process is just like selecting any of the preset or programmable waveforms, and the same rules apply. In this case, it's recommended to put *both* oscillators into editing mode. You can do this quickly by pressing *both* buttons at the same time.

After selecting the edit waveform, expect a brief delay as it loads—the **OCTAVE** display will blink during this process. With the recommended settings and **HOLD** active, you'll hear the waveform build unless it's already been cleared.

Clearing the Edit Waveform

To clear the *edit* waveform before starting:



OSC1 + **OSC2** (together) (**CLEAR EDIT**) then release
Clears the *edit* waveform.

There will be a delay while the waveform clears - if the note was playing, you will hear it disappear. You are now ready to insert or delete harmonics.

Inserting and Deleting Harmonics

When used with keys **1** to **24**, the two waveform buttons have different functions: one is for inserting **INS** and the other for deleting **DEL** harmonics. These functions are labelled below the buttons. In this section, **INS** and **DEL** refer to adding or removing harmonics within waveforms, not to selecting the waveforms themselves.

INS + key 1 to 24

Inserts the harmonic of the selected order (1-24)

- 1** is the fundamental or basic note
- 2** is twice the fundamental frequency
- 3** is three times fundamental frequency
- etc...

DEL + key 1 to 24

Deletes the harmonic of the selected order (if it is present).

Each time a harmonic is inserted or deleted, the waveform is re-constructed. If the note is playing you will hear the new waveform as it develops. (Again the **OCTAVE** display will blink while this happens.)

The fundamental and all harmonics are added to the waveform at the same level, with each able to be entered up to 16 times. The waveform is automatically scaled to full size each time it's calculated, so adding or deleting harmonics only affects their relative size.

This allows for up to 24 harmonics. To expand sound possibilities, this version only scales up to a certain number of harmonic entries; beyond that, further entries cause a *clipping* effect which adds harmonics above the 24th. While these higher harmonics can't be precisely controlled, their influence increases as more entries are added, allowing you to shape the sound as desired.

Similarly, any harmonic can be deleted repeatedly (if it is present) until its level is back down to zero.

Using this repeated insert facility, you can build up the higher harmonics to *unnaturally* high levels and this can produce unusual and interesting waveforms. Note that the fundamental itself does not have to be present; this also contributes to interesting results.



A Guide to Wave-Form Building

These waveforms reveal their full spectrum when played in the lower octave registers. For example, limiting harmonics to lower orders (from the fundamental up to the 10th) creates bell-like sounds.

Using only a few adjacent harmonics, such as 5, 6, and 7, without the fundamental, creates a ring-modulator-like sound. These synthesized ring-mod sounds are typically more consistent across the keyboard compared to traditional ring modulation with rapid *LFO* pitch or filter modulation, which would require the *LFO* frequency to track the keyboard to maintain sound quality.

Using a random mix of harmonics - especially when many high harmonics are set at high levels - produces a sound that is harsher but still not discordant.

The most unusual and interesting sounds come from waveforms with large gaps in their harmonic spectrum. For example, grouping a few harmonics at the low end (like 1, 2, 3), some in the middle (such as 9, 10, 11, entered multiple times to increase relative levels), and several at the high end (like 22, 23, 24, entered multiple times) produces waveforms rarely found in nature, yet still very musical.

Experimenting with different harmonic combinations is the best way to discover new sounds. Using both oscillators together, especially when slightly detuned, creates a much livelier effect than using just one or having both at the same pitch.

Storing the Edited Wave-Form

Once you're satisfied with the edited waveform, you can **store** it in one of the 5 user locations.

OSC1 or OSC2 + STORE + key -8 to -12

stores the *edit* waveform in that *programmable* waveform location ready for selection at any time later on. (The waveform already present will be overwritten.)

You can use *either* waveform button to store a waveform - it doesn't matter which one. Stored waveforms aren't tied to a specific oscillator and can be selected for either one later.

After storing, you remain in edit mode and the edited waveform stays active. To exit edit mode, just select any other waveform as described earlier.

Rapid Waveform Creation

Once you're comfortable with waveform creation, you can speed things up by skipping the edit mode and avoiding the wait after each harmonic change.

This method is identical to normal waveform creation, except you work without hearing the changes. It's faster since intermediate waveform processing doesn't occur, but you won't be able to hear or store the waveform until you enter **EDIT** mode. Follow these steps:

OSC1 + OSC2 then release

ALWAYS clears the edit waveform even if it is not selected on either (**CLEAR EDIT**) oscillator.

INS + key 1 to 24

ALWAYS inserts the chosen harmonic.

DEL + key 1 to 24

ALWAYS deletes the chosen harmonic.

You can enter harmonics rapidly up to their maximum levels (up to 16 times), but you'll need to keep track if you want specific harmonic content. Even mistakes can often lead to interesting and pleasant results.

DON'T try to store the waveform at this stage as it is not fully processed yet.

Finally enter the edit mode (one or both oscillators - it doesn't matter).

OSC1 and/or OSC2 + key 0

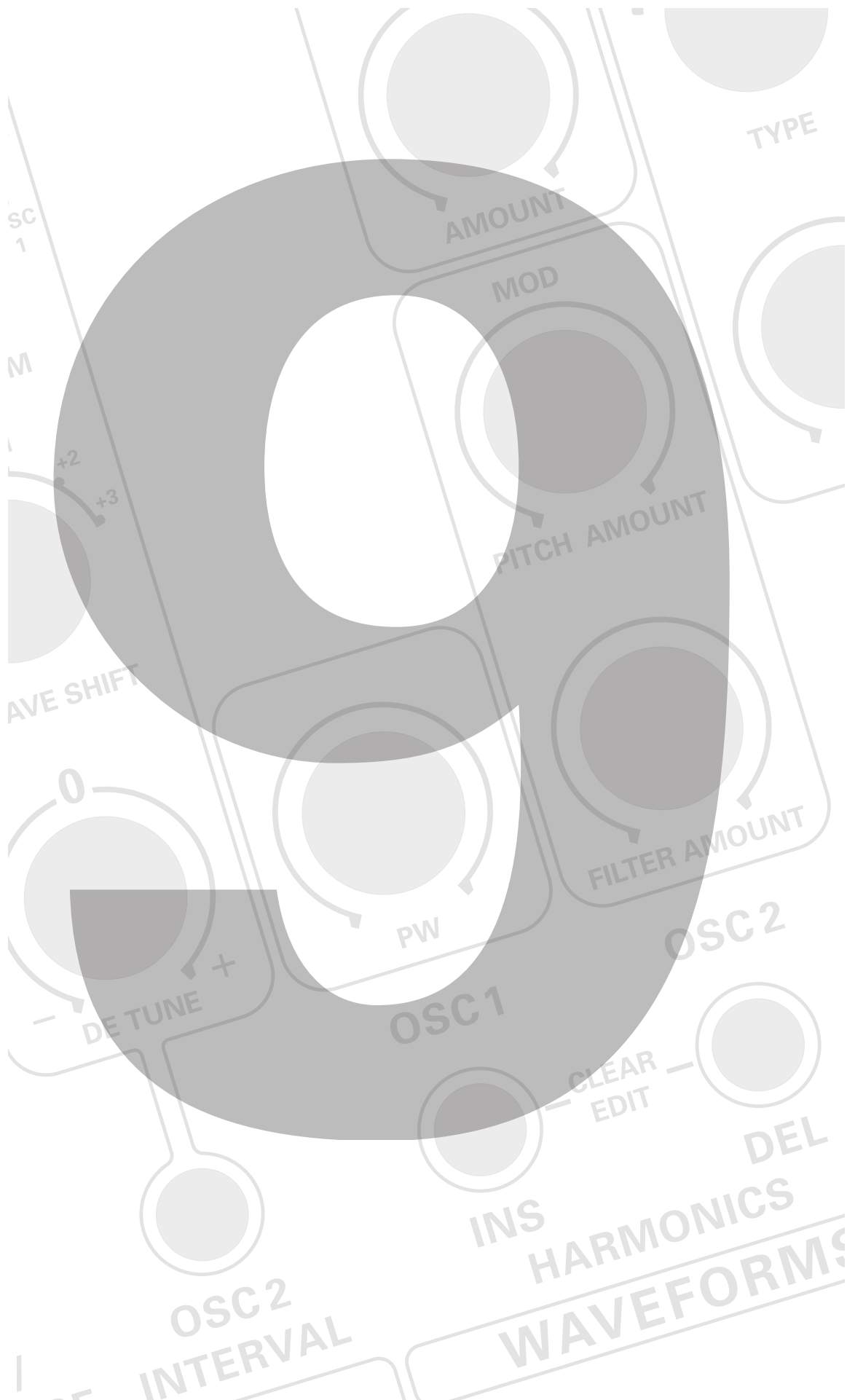
sets up the *WAVEFORM EDIT* mode and the *edit* waveform is constructed.

If the note is playing at the time, it will be heard building up.

Now, the waveform can now be played or stored in the normal way.

See section 9 - *MIDI FUNCTIONS* for permanent storage of waveforms.



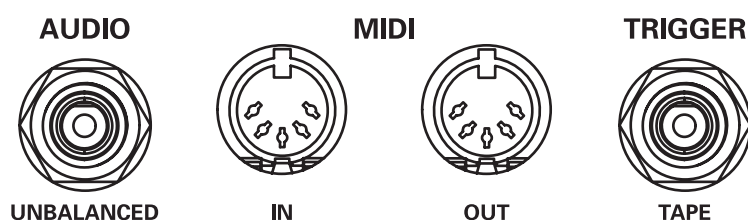


9 MIDI FUNCTIONS

Introduction

The OSCar40 uses the Tauntek MIDI interface and advanced MIDI operating system developed by Bob Grieb (tauntek.com). This new operating system allows for extensive parameter control via MIDI control change messages. *(MIDI CC reception added for all front panel controls except volume.)*

Connections



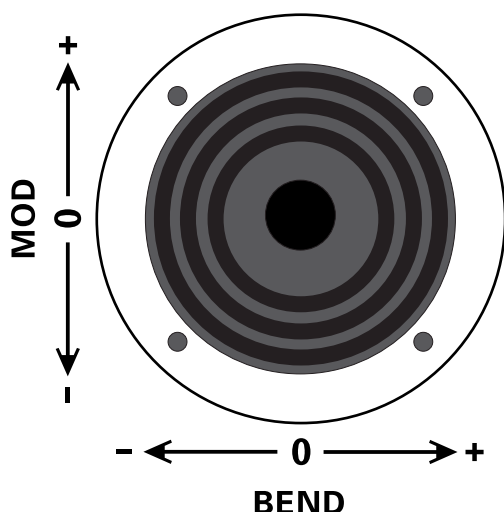
OSCar40 has a MIDI IN for receiving MIDI signals from another instrument and a MIDI OUT for transmitting MIDI signals to another instrument. (Note that signals received at MIDI IN will not be re-transmitted from MIDI OUT on this instrument. The OSCar40 has no MIDI THRU function)

Use standard MIDI cables for interconnection between instruments. The simplest connection arrangement is MIDI OUT of one instrument (the master) to MIDI IN of another instrument (the slave).



Selectable Control Options

The MIDI system in the OSCar40 offers eight control options. By default some of these are enabled and some are disabled. Holding down the **SPACE** push-button on the joystick panel causes the **OCTAVE** display to show temporarily whether 5 of these options are enabled and allows all of the options to be changed using the keyboard. Changes to MIDI channel and MIDI settings preserved when power is off.



1) Performance Joystick Option

Enabled on power-up. Transmit and receive joystick positions. Accessed by holding down the **SPACE** button and then pressing key -**11** of the keyboard. While the space button is held down, the *first* (left most) LED of the **OCTAVE** display will light if wheel signals are enabled. Repeating the procedure simply puts the option into the other state i.e.. off if it was previously on - on if it was previously off.

Modulation signals do not normally allow for the OSCar40's negative modulation; the OSCar40 therefore transmits a normal (positive) modulation wheel signal whether it is turned up or down. The scale factor of modulation wheel signals has not been universally agreed between manufactures but the OSCar40 is fully compatible with most instruments in this respect and at least usable with others.

2) Program Change Option

Disabled on power-up. Transmit and receive program (voice) changes. Access this option the same way as the joystick option, but press **SPACE** and key **-9**. If program changes are enabled, the *second* LED from the left will light while you hold the space button. Repeat the combination to toggle ON/OFF.

The OSCar40 supports 36 voice programs, so it can only transmit or receive the first 36 of 128 possible MIDI programs.

The order of program numbers on the OSCar40 is **-12** to **-2** then **1** to **24** corresponding to program change numbers **1-36**.

Note: MIDI Bank change is *not* supported.

3) Key ON/OFF Option

Enabled on power-up. Transmit and receive key-on and key-off signals. Press **SPACE** and key **-6** to access this option. If keyboard signals are enabled, the *third* LED from the left will light while you hold the space button. Repeat the combination to toggle ON/OFF.

Notes received by the Oscar40 outside the range from G - more than 2 octaves below key 0 to F - more than 2 octaves above key 0 will be transposed by whole octaves so that they fall within this range.



4) Sequencer and Arpeggiator Option

Disabled on power-up. Transmit sequencer and arpeggiator MIDI notes. To enable sequencer or arpeggiator MIDI note out, press **SPACE** and key **-4**. The *fourth* LED from the left will light if these signals are enabled. Repeat the combination to toggle ON/OFF.

These are transmitted in addition to keyboard signals, if those are also enabled.

This feature lets you use the OSCar40 as a monophonic MIDI sequencer or arpeggiator for *any* MIDI slave instrument. With keyboard signals enabled, you can play live over the sequence on a polyphonic slave. The **GATE TIME** control adjusts MIDI note duration as usual.

Note: When the sequencer is running, the arpeggiator is disabled.

5) Omni Mode Option

Enabled on power-up. Accessed as before but using **SPACE** and key **-2**. The *fifth* LED will light if the omni mode is selected.

When in OMNI mode, the OSCar40 will receive signals on any of the 16 MIDI channels. When the OMNI mode is switched off, signals will only be received through the MIDI channel *selected* (see next section). Transmission is always on the selected MIDI channel.

Control options with no status display:

6) Timing Clock Option

Disabled on power-up. Accessed using **SPACE** + key **20**.
Repeat the combination to toggle ON/OFF.

The MIDI timing clock works like external trigger pulses. It can also be automatically controlled by MIDI Start and Stop messages, which most drum machines and sequencers send at the beginning or end of a sequence. The Start message enables the OSCar40 timing clock; the Stop message disables it and resets the sequence. This ensures full synchronisation and control. The standard clock rate is 24 clocks per quarter note, and setting External Trigger Divide to 12 is usually best.

7) Pull-off Note Option

On power-up, pull-off response is enabled. To switch it off, hold down the **SPACE** button and press key **22**.
Repeat the combination to toggle ON/OFF.

Because the OSCar40 is monophonic, it normally uses *pull-off* playing - sounding a new note when a key is released while another is held, which is ideal for monophonic lines. However, when layering with a polyphonic instrument, you may want to disable *pull-off* so notes only respond to new key presses.

8) MIDI Channel Assignment

Set to channel 1 on power up, changed by holding down **SPACE** and pressing key **1-16** to select the channel of that number.

MIDI is always sent on the *selected* channel. With OMNI mode **OFF**, only signals on the selected channel are *received* - this acts as a channel assignment mode. With omni mode **ON**, the OSCar40 accepts MIDI signals on *any* of the 16 channels.

The ALL NOTES OFF Message

Pressing and releasing the **SPACE** button turns off *all* notes, no matter how they were activated. If keyboard signals are enabled, it also sends an *All Notes Off* message to connected MIDI devices.



MIDI CC Dump

CC DUMP



MIDI **CC DUMP** sends every parameter (with the exception of **VOLUME**) each time a voice is loaded. This can be enable or disable. Disable by default.

This is a useful feature for integration with external editors.

TUNE/TRANPOSE + key **OFF (9)** then release
disables the **CC DUMP**

TUNE/TRANPOSE + key **ON (10)** then release
enables a **CC DUMP** each time a new voice is loaded

Summary of MIDI Options

Accessed by holding down **SPACE**:

Function	Button	Key	Power On Status	Octave display when enabled
Joystick	SPACE	-11	Enabled	
Program Change	SPACE	-9	Disabled	
Key ON/OFF	SPACE	-6	Enabled	
Seq/Arp Transmit	SPACE	-4	Disabled	
Omni Mode	SPACE	-2	ON	
Timing Clock	SPACE	20	Disabled	X
Pull-off Response	SPACE	22	Enabled	X
Channel Select	SPACE	1-16	OMNI	X
All Notes OFF	SPACE			X
CC Dump OFF	TUNE	9	OFF	X
CC Dump ON	TUNE	10	OFF	X

SYSEX Data Transfer

Using the MIDI SYSEX Exclusive format, programmable **WAVEFORM**, **VOICE** (*as the entire working memory bank*) and **SEQUENCE** data can be saved as SYSEX files or transferred to another Oscar40. It's generally good practice to save all types of data together in one file, that way all the voices, waveforms and sequence data is collected conveniently together.

INSERT + WAVEFORM buttons

selects custom waveform data

LED 1 on the **OCTAVE** display will be lit

INSERT + VOICE buttons

selects the entire working memory voice bank

LED 2 on the **OCTAVE** display will be lit

INSERT + SEQ buttons

selects the sequencer data

LED 2 on the **OCTAVE** display will be lit

INSERT + STORE buttons

selects **ALL** data types, voices, waveforms and sequence data

LEDs 1, 2 and 3 on the **OCTAVE** display will be lit

To send the selected data as SYSEX press the **SPACE** button and the data will be transmitted. When saving **ALL** data types at the same time, the OSCar40 will send each data type as a separate set of data with a short pause between each type. When sending to a computer program for storage, you may need to configure the receiving application in order to successfully record an **ALL** data SYSEX file.

For example, if you are using snoize.com's *SysEx Librarian* on MacOS, you must use the *Record Many* function to successfully record **ALL** the data types as one SYSEX file. Using *Record One* will result in *only* the Waveform data being recorded.

MIDI Control Change

Every parameter of the OSCar40 (with the exception of the **VOLUME** control) can be controlled via MIDI control change messages allowing for complete automation of the OSCar40 from a DAW (*digital audio workstation*) or MIDI controller. Refer to your application/controller user manual for instructions on how to send MIDI CC messages. A full list of MIDI CC messages and their data ranges is provided on the next pages.



MIDI CC Message List

Function	CC #	CC Range	Parameter Value	Comment
Osc1 Waveform	14	0-127	0-15	
Osc2 Waveform	15	0-127	0-15	
Pulse Width	16	0-127	1-254	
Osc2 Detune Amt	17	0-127	1-254	
Osc2 Octave Shift	18	0-127	0-5	
Osc2 Semitone Offset	19	0-127	0-14	
Tuning Semitone	20	0-127	0-14	
Octave setting	21	0-127	2-6	
Glide Type	22	0-127	0-5	
Glide Rate	23	0-127	1-254	
Direct Pitch Mod Amt	24	0-127	1-254	
Mod Wheel Pitch Amt	25	0-127	1-254	
Bend Amt	26	0-127	1-254	
Oscillator Balance	27	0-127	1-254	
Noise Balance	28	0-127	1-254	
Filter Driver	29	0-127	1-254	
Filter Frequency	30	0-127	1-254	
Filter Separation	31	0-127	1-254	
Filter Resonance/Q	32/71	0-127	1-254	
Filter Type	33	0-127	0-5	
Filter Attack	34	0-127	1-254	
Filter Decay	35	0-127	1-254	
Filter Sustain	36	0-127	1-254	
Filter Release	37	0-127	1-254	
Direct Filter Mod Amt	38	0-127	1-254	
Env Filter Mod Amt	39	0-127	1-254	
Mod Wheel Filter Amt	40	0-127	1-254	
Volume Attack	41/73	0-127	1-254	
Volume Decay	42	0-127	1-254	
Volume Sustain	43	0-127	1-254	
Volume Release	44/72	0-127	1-254	
Gate Time	45	0-127	1-254	
LFO Waveform	46	0-127	0-5	
LFO Rate	47	0-127	1-254	
Intro Delay Time	48	0-127	1-254	
Trigger Type	49	0-127	0-5	
Function	50	0-127	0-5	
Arpeggiator Mode	51	0-127	0-15	(some values not supported)
LFO to Osc1 disable	52	0-127	0 or 1	
LFO hard sync enable	53	0-127	0 or 1	
Tempo	54	0-127	1-254	
PWM mod rate	55	0-127	0-3	(0 is slowest)



CC values needed to control the OSCar40:		CC values sent in patch dump:	
<i>Note: "Value" is the actual value in the OSCar40's RAM, so this is what would be in a patch dump. Number range is what needs to be sent to the synth using MIDI.</i>		<i>Note: Only the lower CC numbers will be sent for the four parameters that have two CC's assigned to them for reception.</i>	
Parameters that have two settings:			
0-63 Value is 0		Parm = 0, 0 sent	
64-127 Value is 1		Parm = 1, 64 sent	
Parameters that have four settings:			
0-31 Value is 0		Parm = 0, 16 sent	
32-63 Value is 1		Parm = 1, 48 sent	
64-95 Value is 2		Parm = 2, 80 sent	
96-127 Value is 3		Parm = 3, 112 sent	
Parameters that have 5 settings:			
0-24 First value		First parm value, 12 sent	
25-49 Second value		Second parm value, 37 sent	
50-74 Third value		Third parm value, 62 sent	
75-99 Fourth value		Fourth parm value, 87 sent	
100-127 Fifth value		Fifth parm value, 102 sent	
Parameters that have 6 settings:			
0-20 First value		First value, 10 sent	
21-41 Second value		Second value, 31 sent	
42-62 Third value		Third value, 52 sent	
63-83 Fourth value		Fourth value, 73 sent	
84-104 Fifth value		Fifth value, 94 sent	
105-127 Sixth value		Sixth value, 115 sent	
Parameters that have 15 or 16 settings:			
0-7 First value		First value, 4 sent	
8-15 Second value		Second value, 12 sent	
16-23 Third value		Third value, 20 sent	
24-31 Fourth value		Fourth value, 28 sent	
32-39 Fifth value		Fifth value, 36 sent	
40-47 Sixth value		Sixth value, 44 sent	
48-55 Seventh value		Seventh value, 52 sent	
56-63 Eighth value		Eighth value, 60 sent	
64-71 Ninth value		Ninth value, 68 sent	
72-79 Tenth value		Tenth value, 76 sent	
80-87 Eleventh value		Eleventh value, 84 sent	
88-95 Twelfth value		Twelfth value, 92 sent	
96-103 Thirteenth value		Thirteenth value, 100 sent	
104-111 Fourteenth value		Fourteenth value, 108 sent	
112-119 Fifteenth value		Fifteenth value, 116 sent	
120-127 16th value if there is one, otherwise, the 15th		Sixteenth value, 124 sent	
Semitone parameters settings:			
Parm = 0, 116 sent	+7		
Parm = 1, 108 sent	+6	Parm = 8, 52 sent	-1
Parm = 2, 100 sent	+5	Parm = 9, 44 sent	-2
Parm = 3, 92 sent	+4	Parm = 10, 36 sent	-3
Parm = 4, 84 sent	+3	Parm = 11, 28 sent	-4
Parm = 5, 76 sent	+2	Parm = 12, 20 sent	-5
Parm = 6, 68 sent	+1	Parm = 13, 12 sent	-6
Parm = 7, 60 sent (0 offset)	0	Parm = 14, 4 sent (-7)	-7

